《材料物理》



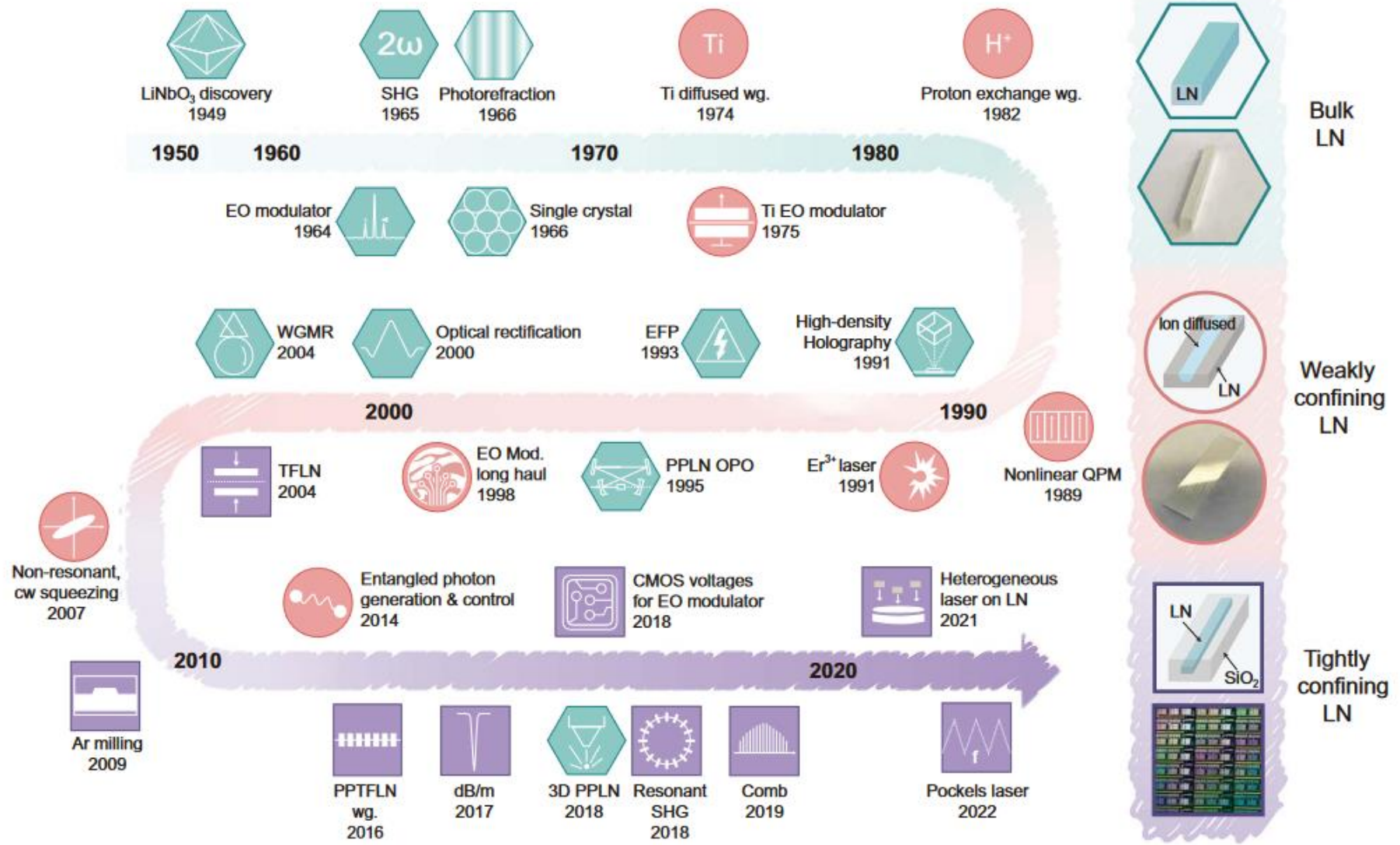
铌酸锂晶体

孔勇发

主要内容

- 研究背景
- 基本习性
- 掺杂工程
- 畴工程
- 近化学计量比晶体
- 结语

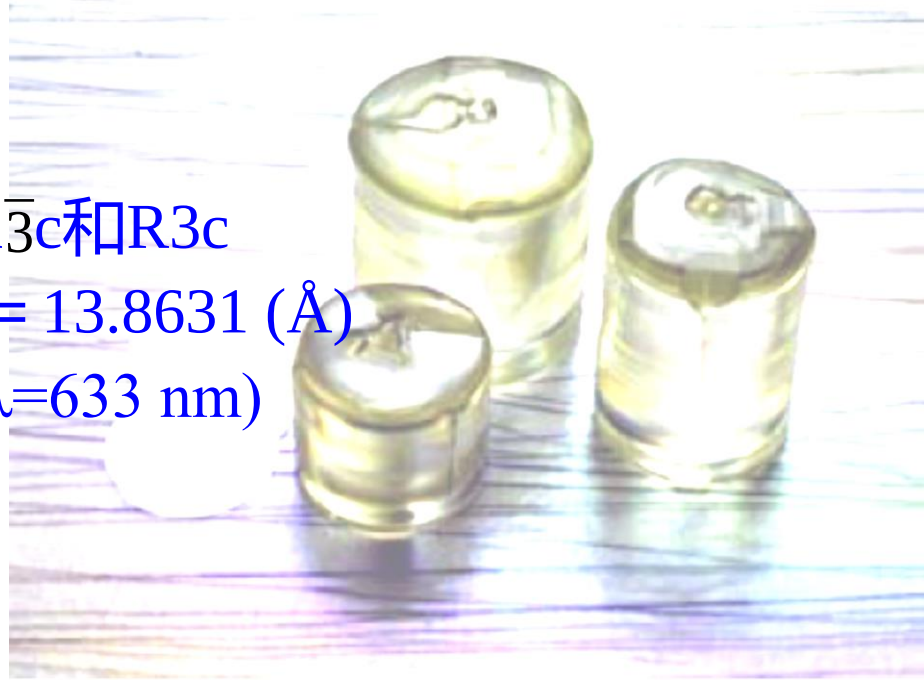
铌酸锂的研究历程



Science 379, eabj4396 (2023).

一、研究背景

- 铌酸锂（$LiNbO_3$；LN）
- 1965年Ballman首先用提拉法生长出铌酸锂单晶。
- 目前已知的居里点最高（1210°C）、自发极化最大（0.71 C/m^2）的铁电体。
- 晶体密度：4.612 (g/cm^3)
- 莫氏硬度：5
- 熔点：1260 (°C)
- 顺电相和铁电相的空间群：$R\bar{3}c$和R3c
- 晶胞参数：a = 5.1483 (Å)；c = 13.8631 (Å)
- 折射率：n_o=2.286, n_e=2.202 (λ=633 nm)
- ……



铌酸锂晶体的特性

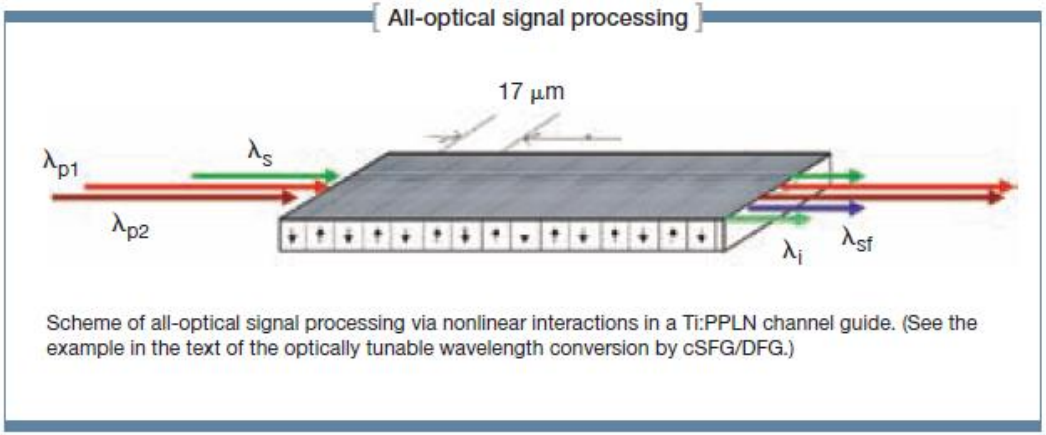
- **多效应**：集电光、声光、压电、光弹、非线性、光折变及激光活性等效应于一身。
- **多用途**：光波导、调制器、隔离器、频率转换、光参量振荡、窄带滤波器、传感器、光陀螺、海量存储等方面，具有十分广阔的市场应用前景。
- **性能的可调控性**：对多种杂质具有良好的固溶性，且晶体性能随杂质及组分的变化而明显变化。
- 热、化学、机械**稳定性**；易于生长大尺寸晶体；成本低等。

“Optical silicon”

New materials renew life for integrated optics

Lawrence Gasman
WDM Solutions, November, 2001

Material systems based on silica on silicon, gallium arsenide, **lithium niobate**, and indium phosphide are contenders for the role of "**optical silicon**."



Scheme of all-optical signal processing via nonlinear interactions in a Ti:PPLN channel guide. (See the example in the text of the optically tunable wavelength conversion by cSFG/DFG.)

nature publishing group

Materials Update: Material of the month

November 2002: Lithium niobate

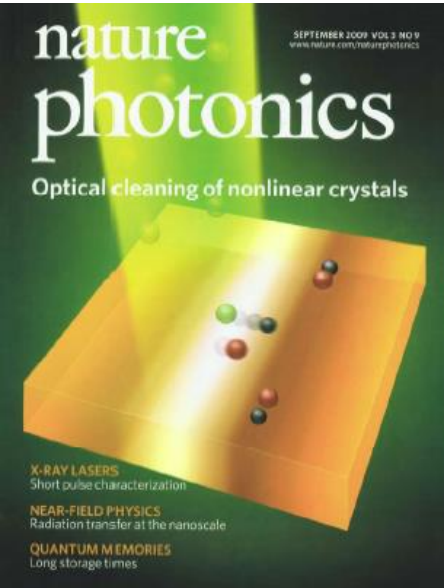
- In the field of nonlinear optics there have been many contenders for the title of **all-star material of the world**.
- But for day-to-day applications, the most successful of these nonlinear materials is lithium niobate.
- Indeed, because of its availability, widespread use and versatility, it has been dubbed by many as the "**silicon of nonlinear optics**".

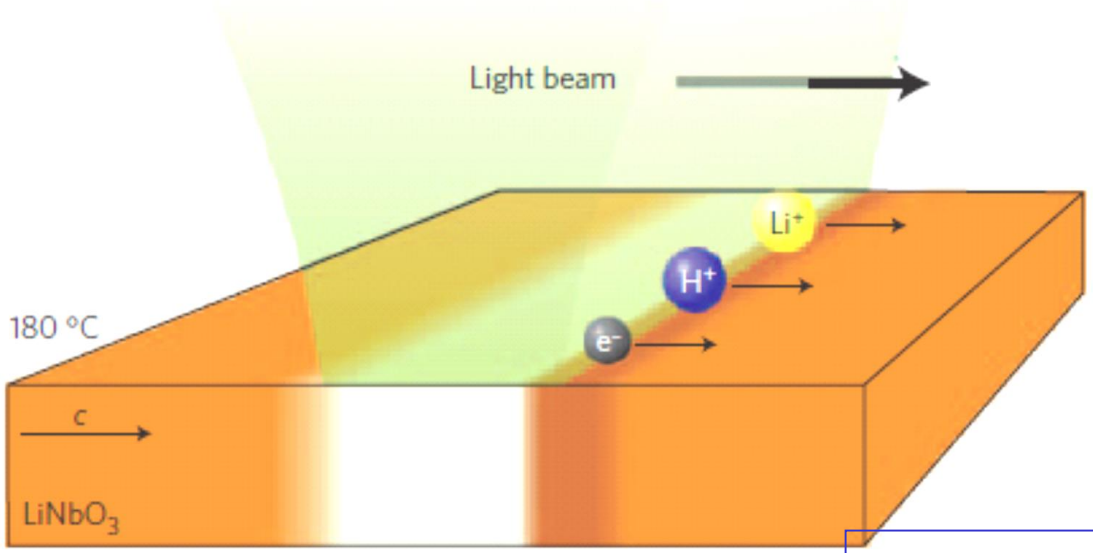


LETTERS
PUBLISHED ONLINE: 23 AUGUST 2009 | DOI: 10.1038/NPHOTON.2009.142
nature photonics

Silicon of photonics

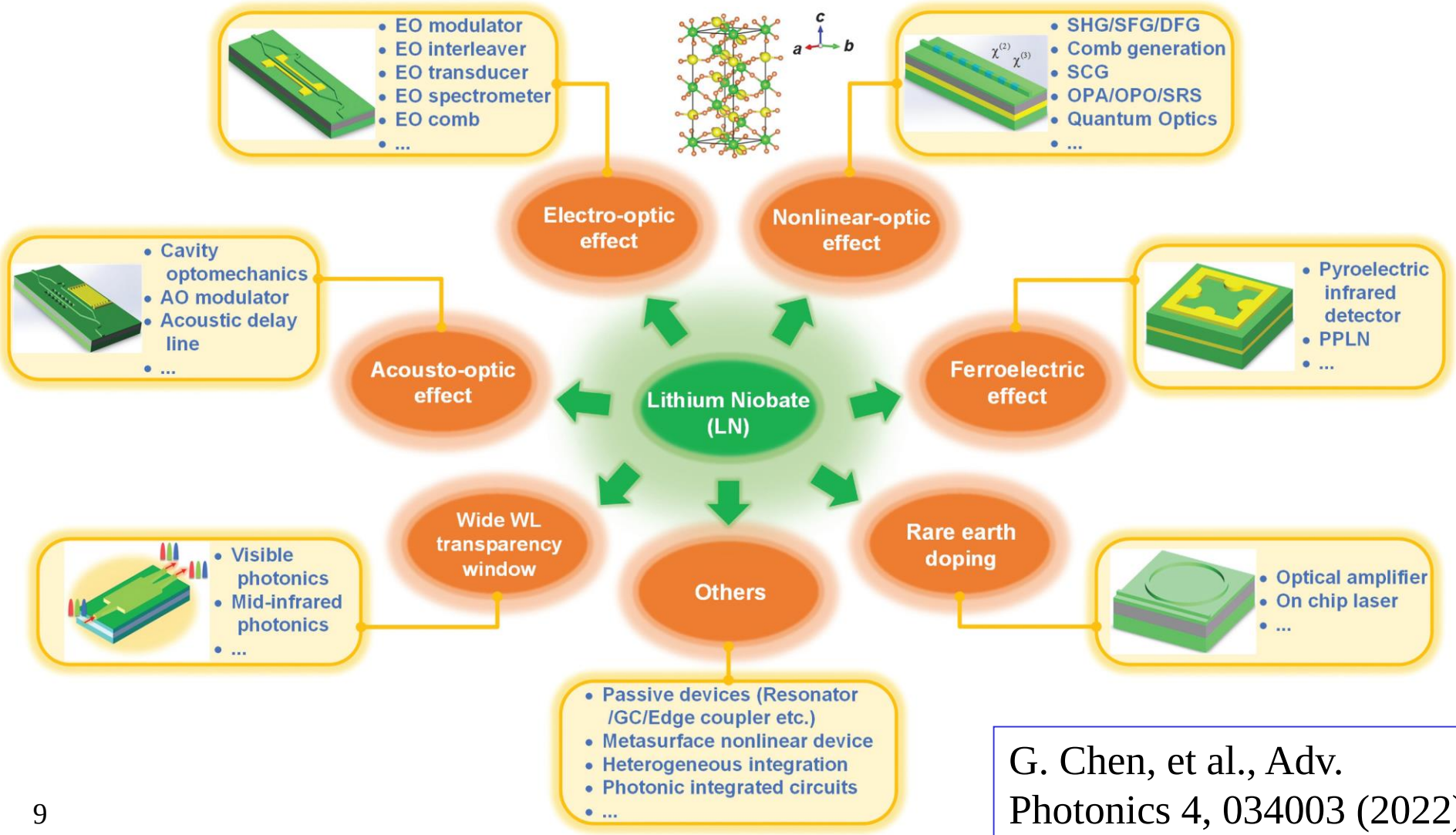
- Lithium niobate ($LiNbO_3$), also called the 'silicon of photonics', is indispensable in advanced photonics and nonlinear optics.





M. Kösters1, et al., Nature Photonics 3, 510 (2009)

Overview of LN photonics



G. Chen, et al., Adv. Photonics 4, 034003 (2022)

Information System

信号源

⇒调制

⇒放大

⇒传输

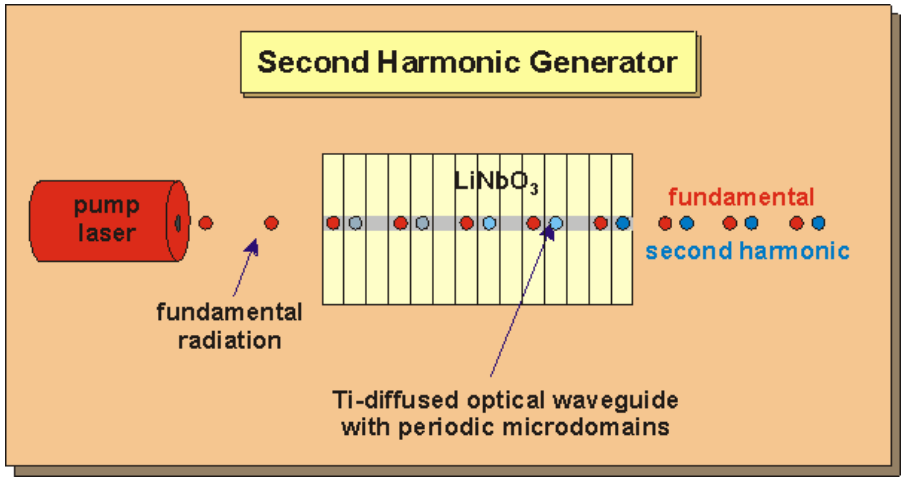
⇒接收

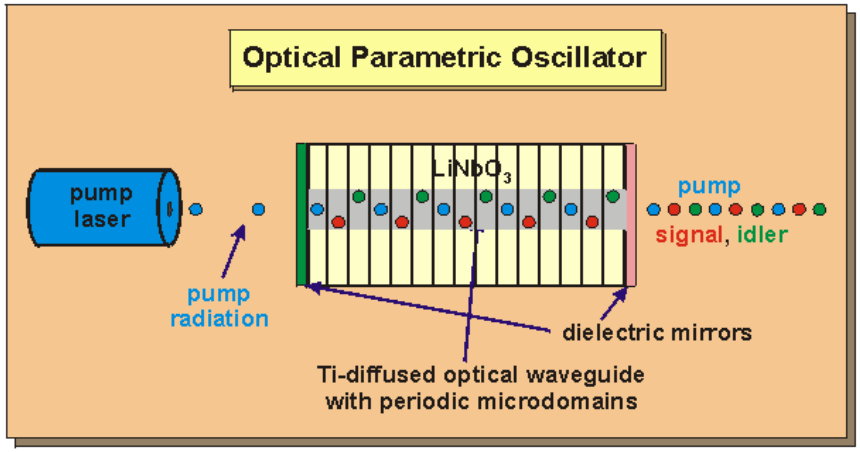
⇒解调

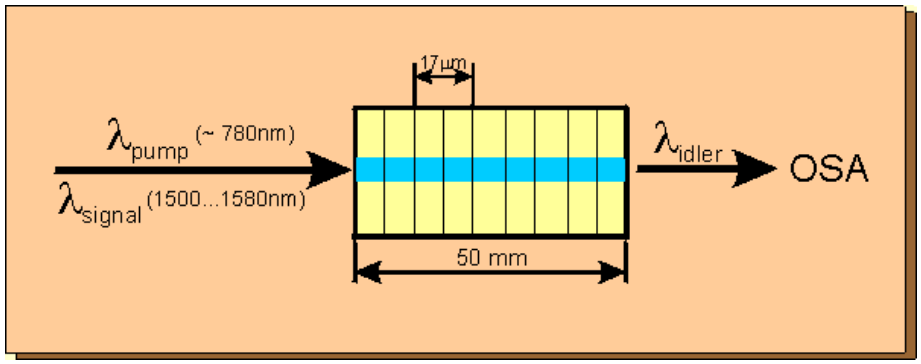
⇒ 输出

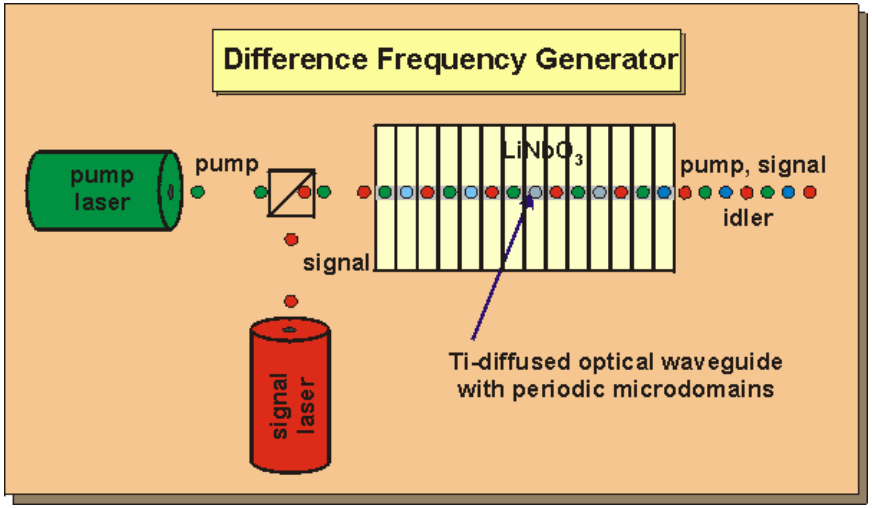
（显示，打印，存储等）

Lasers

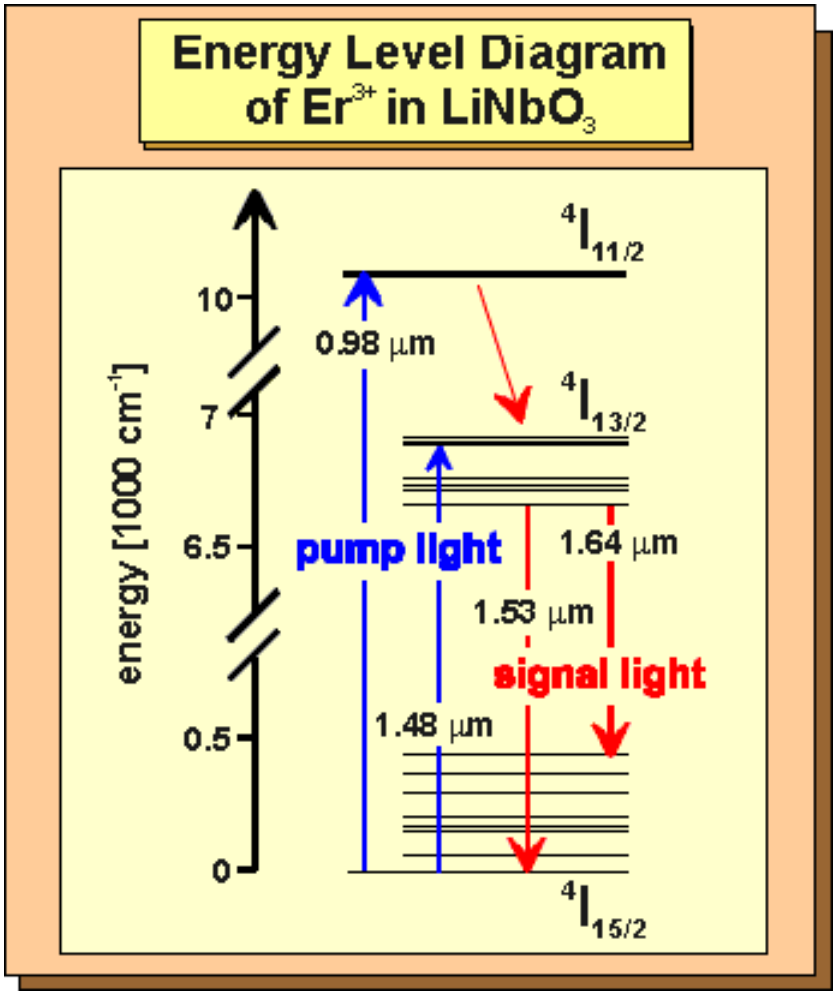

Second Harmonic Generator
pump laser
LiNbO3
fundamental
second harmonic
fundamental radiation
Ti-diffused optical waveguide with periodic microdomains


Optical Parametric Oscillator
pump laser
LiNbO3
pump
signal, idler
pump radiation
dielectric mirrors
Ti-diffused optical waveguide with periodic microdomains


17µm
λpump (~ 780nm)
λsignal (1500...1580nm)
λidler
OSA
50 mm

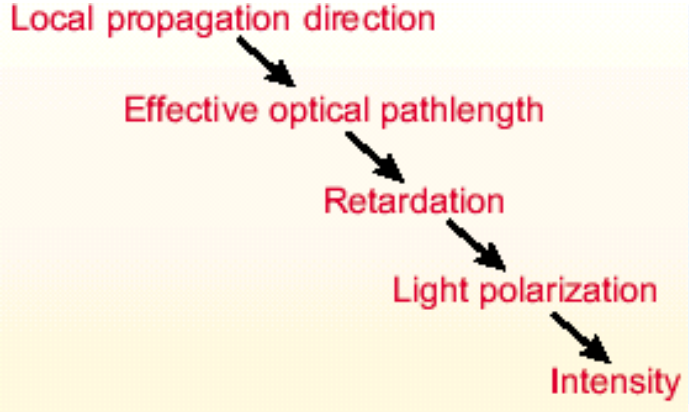

Difference Frequency Generator
pump laser
pump
LiNbO3
pump, signal
idler
signal
signal laser
Ti-diffused optical waveguide with periodic microdomains

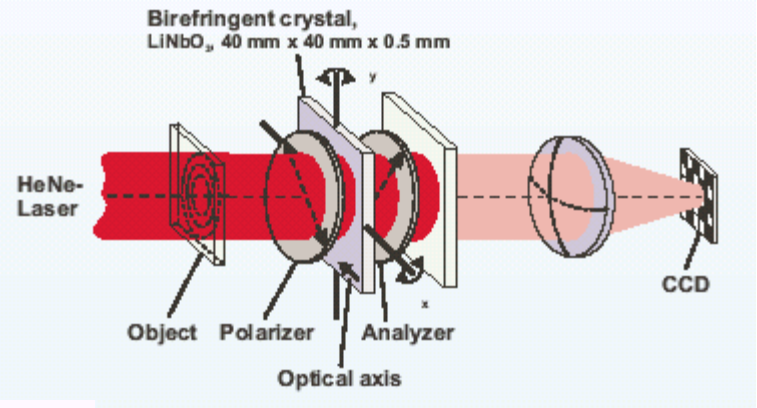
Lasers

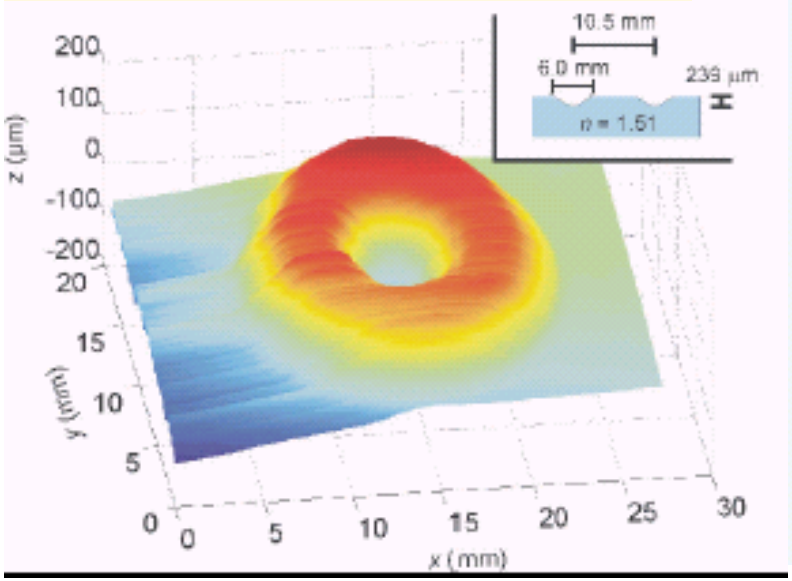


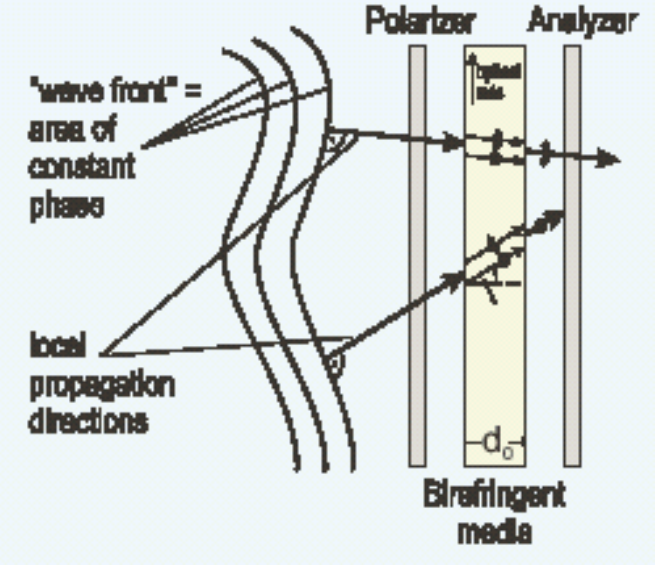
- Modelocked lasers
- Q-switched lasers
- Tunable lasers
- DBR-lasers
- THz source;
- Quantum-entangled photon source

Wave-Front Sensor







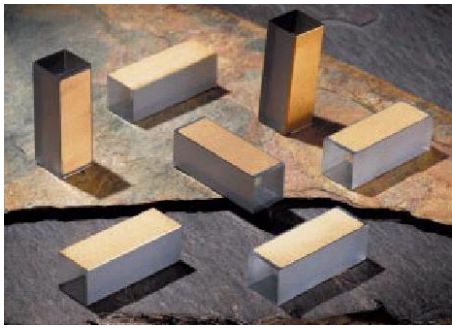


Electro-Optic Q-switch crystal

SPECIFICATIONS

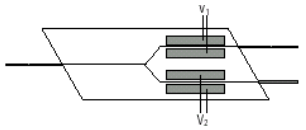
Orientation	Sides parallel to z-axis to ±10 minutes of arc
Polish	z-faces flat to λ/8 and parallel to 10 seconds of arc
Finish	x- and y-faces fine ground and all edges chamfered
Electrodes	X-faces, gold chromium electrodes
Coatings	AR coated with ≤0.2% reflection/surface @ 1064 microns
Optical Surface Quality	20/10 scratch dig
Wavefront Distortion	≤ λ/4, single pass @ 633 mm
Extinction Ratio	
Passive	≥100:1 @ 633 nm
Dynamic	≥100:1 @ 633 nm between crossed polarizers ≥50:1 @ 633 nm between parallel polarizes
Optical Damage Threshold	>200 MW/cm^2 * @ 1.064 μm

* Optical damage thresholds of >500 MW/cm^2 are available



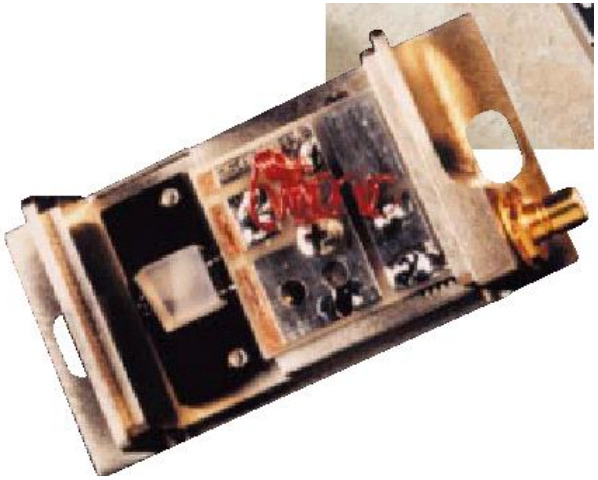
Electro-Optic Modulators

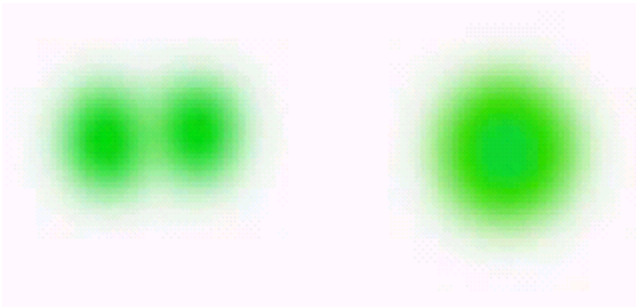
lithium niobate is the only material used for 1300 nm and 1550 nm.

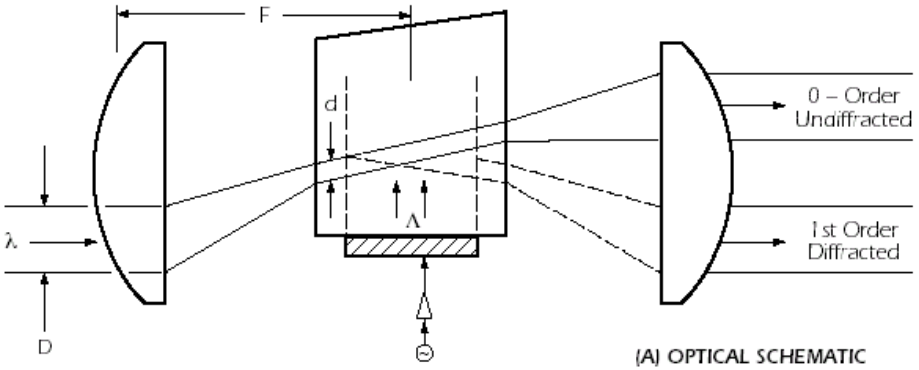


- Waveguides are made by diffusing titanium or hydrogen into the **lithium niobate**, increasing the refractive index of a narrow stripe waveguide.
- Widely used for modulating signal speed > 10 Gbit/s for both digital and analog transmissions.

Acousto-Optic modulator





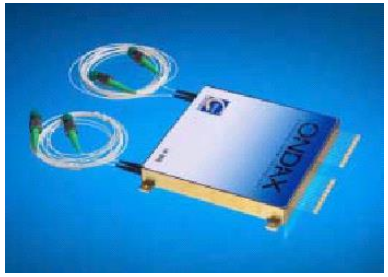


(A) OPTICAL SCHEMATIC

DWDM Filters

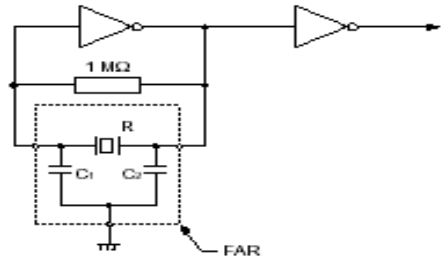
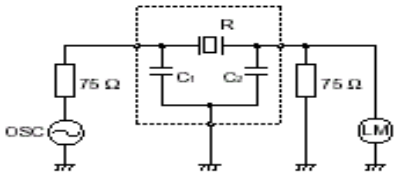


	2-Port			3-Port *			
Specification	**Min.**	**Typ.**	**Max.**	**Min.**	**Typ.**	**Max.**	**Units**
Bandwidth @ -0.5 dB	20	25	38	20	25	38	GHz
Adjacent channel isolation	15	20	30	15	20	30	dB
Non-adjacent channel isolation	20	25	40	20	25	40	dB
Drop loss		1.5	2.0		1.5	2.0	dB
PDL		0.2	0.3		0.2	0.3	dB
Insertion loss		N.A.				2.5	dB
Tuning time		2			2		s
Number of channels	32			32			
Operating case temp. range	15		40	15		40	^{o}C
Non-operating temp. range	0		70	0		70	^{o}C
Crosstalk					25		dB
Power consumption		5			5		W
Size	120 x 120 x 18						mm



Piezoelectric Resonator(4 to 16 MHz)

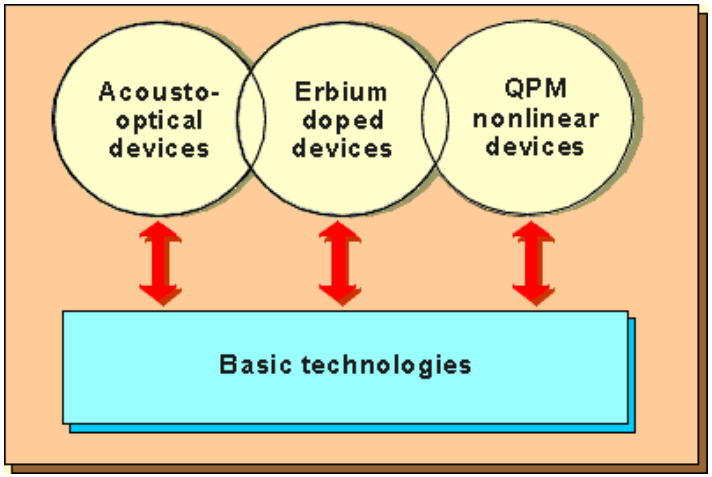


Parameter \ Series	C4 series (P/Q type)	Remarks
Material	Lithium Niobate ($LiNbO_3$)	
Frequency	4 to 16 MHz	
Standard frequency	See "■ Standard Frequency."	
Initial frequency deviation	±0.5% (M)	The ±0.3% (K) version can also be produced.
Temperature characteristic (–40°C to +105°C)	$\pm^{0.9}_{1.0}$% (M)	Reference temperature: +25°C
Capacity of built-in capacitor	20±8 pF (standard)	10±4 pF, 30±8 pF are also available. Capacity is specified by Fujitsu, considering matching data with applied IC (mainly microcomputer).
Aging stability	Within ±0.1%	For ten years at room temperature
PCT	96 hours guaranteed	Unsturated PCT: 121°C 2 atomospheric pressures
Operating temperature	–40°C to +105°C	
Storage temperature	–55°C to +105°C	
Standard measuring circuit	• Resonant frequency  4 MHz to 10 MHz IC: MB84069B×2 10 MHz to 16.0 MHz IC: MC74HC04×2 • V_{DC}: 5 V DC • R: Resonator • C_1, C_2: Loading capacitors (built-in) • Serial resonant resistance  R: Resonator Measuring instrument: Network analyzer	

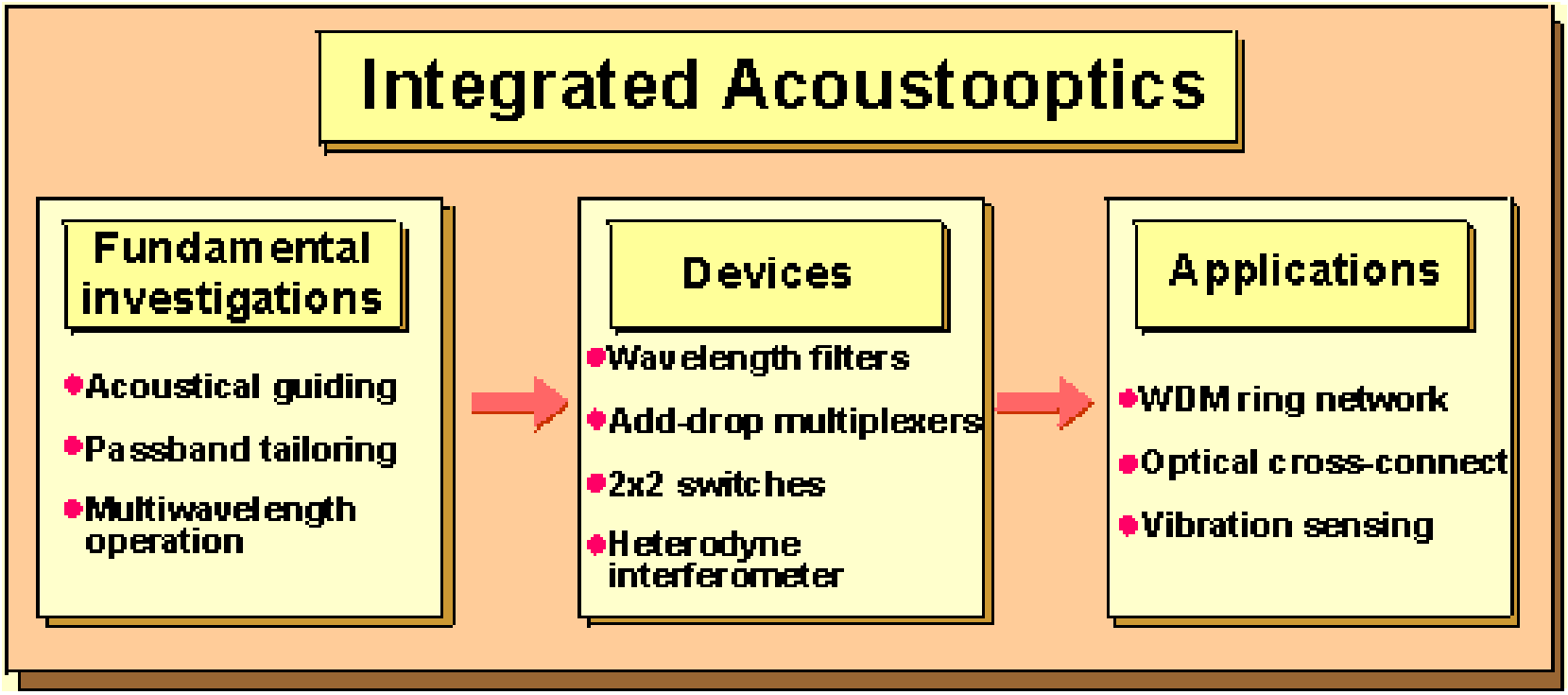
Integrated Optics In $LiNbO_3$

Integrated optical circuits can be realized by two different approaches:

- Hybrid integration: several devices are fabricated on different materials, with each optimized in a given material, and combined on a common substrate.
- Monolithic integration: all devices are fabricated on a common substrate.($LiNbO_3$)__mass production can be achieved by automatic batch processing,



Integrated Acoustooptics in Lithium Niobate



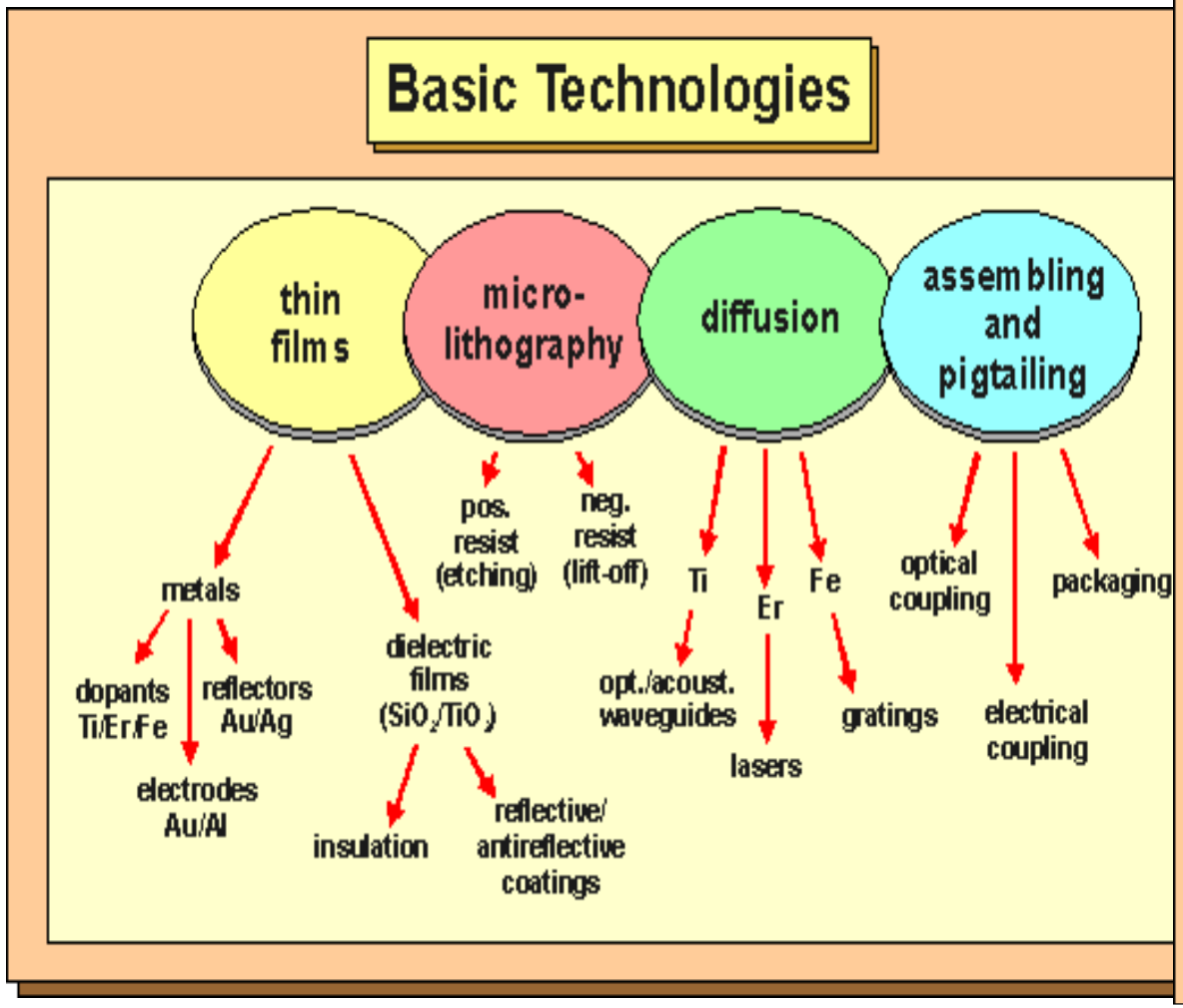
optical waveguide substrates

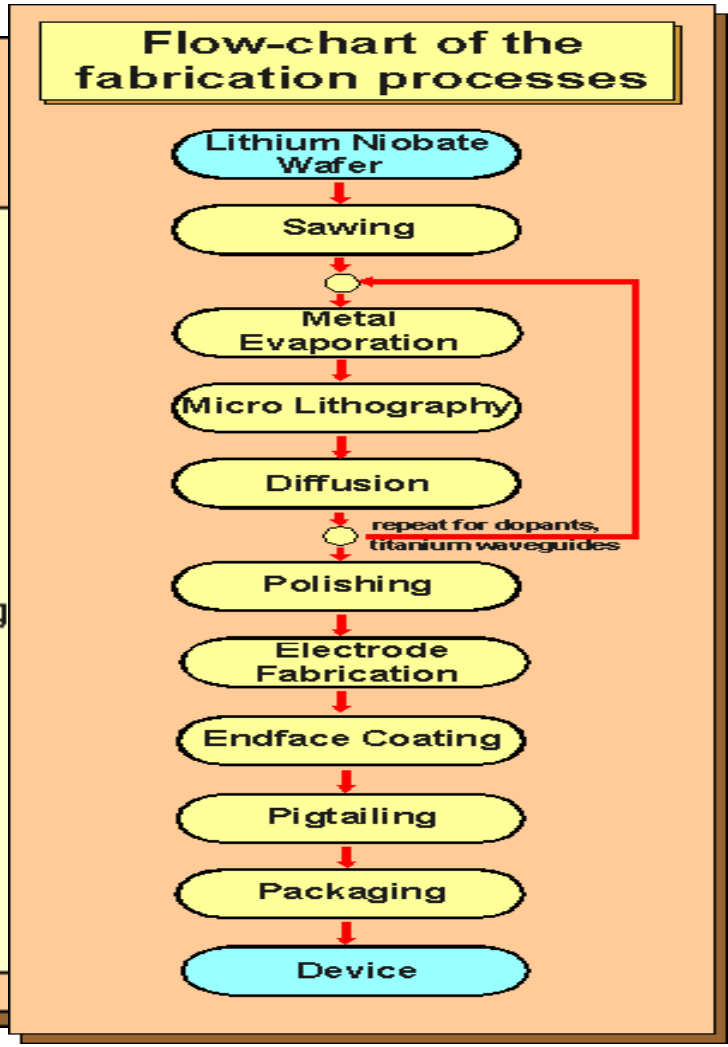
SPECIFICATIONS

Composition	Congruent within 0.01 mol% Li_2O; typical compositional uniformity is better than ±0.005 mol% Li_2O
Curie Temperature	1142.3 ±0.7°C by the dielectric technique
Purity	Typically less than 2 ppm for each transition metal (including iron) in starting chemicals and in grown chemicals
Nominal Birefringence	$n_o - n_e = 0.085 \pm 0.001$
Birefringence Uniformity	Typically ±0.00007
Scatter	Typically free of scatter centers under optical microscopy
Axes Orientation	±30 arc minutes; see reverse side for orientation of reference flats.
Surface Finish	All faces to a 10/5 scratch/dig finish per MIL-O-13830A. Flatness for the 100 mm diameter wafers is 15 microns T.T.V. while the 76.2 mm wafers are flat to 10 microns T.T.V.



Basic Technology

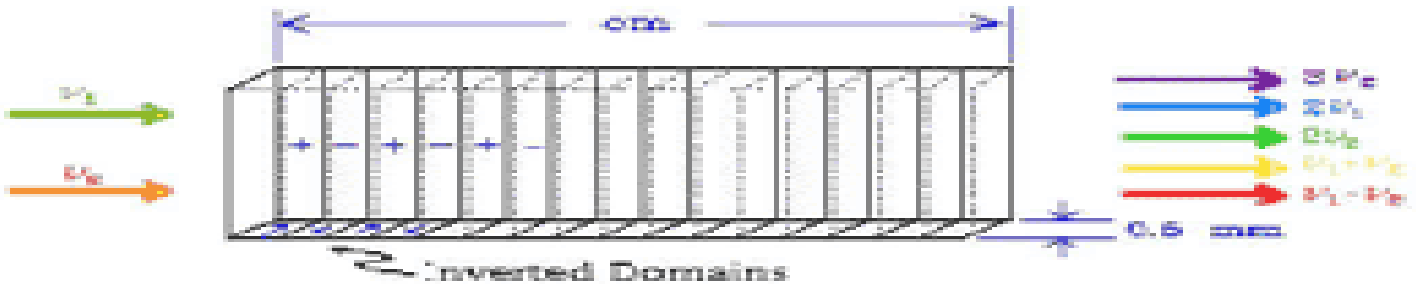




Large all-colour Display

Extended Wavelength Coverage with nonlinear mixing in

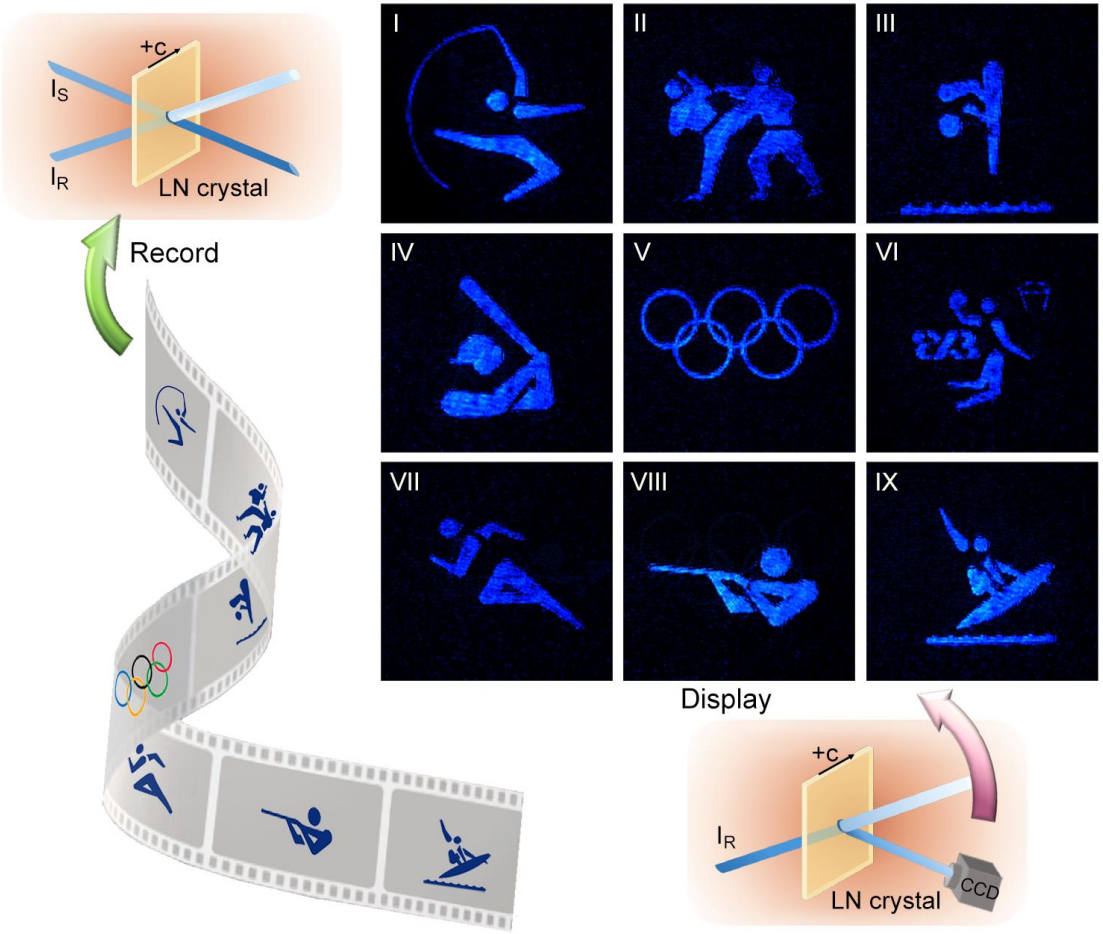
Periodically Poled Lithium Niobate

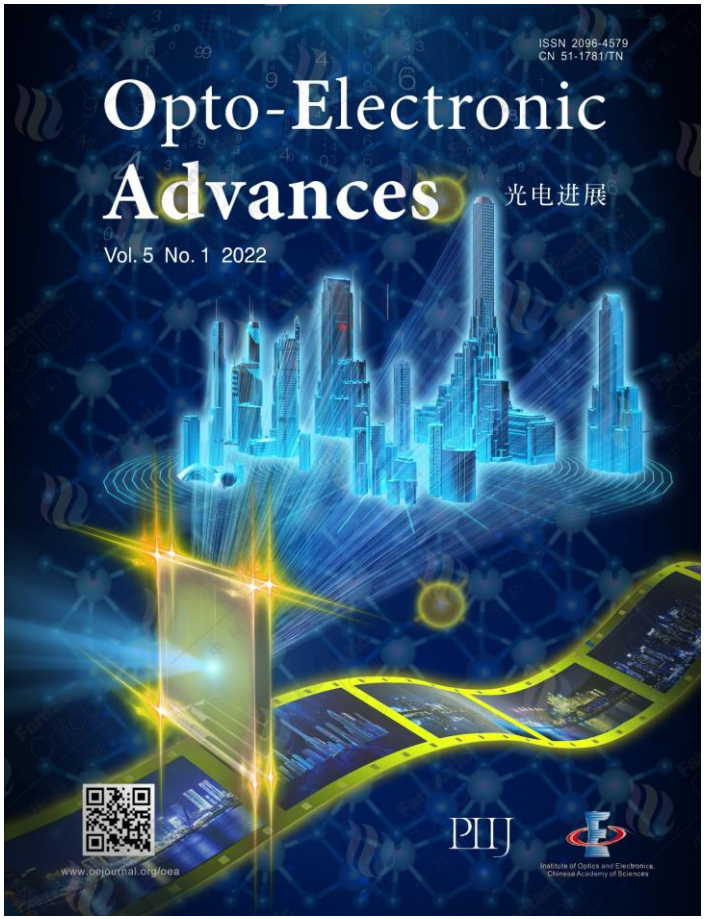


* $P_{om} \propto P_1 P_2$ (10^{-8}/watt) Even single pass through crystal

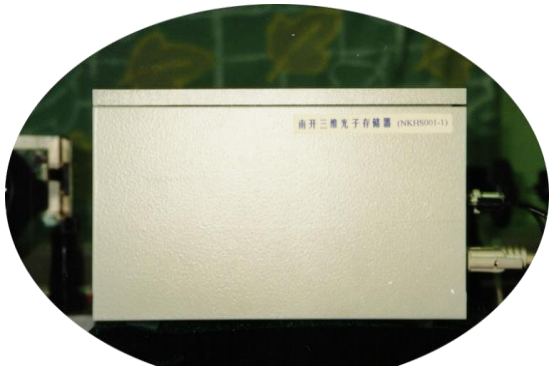
* Useful from Ultraviolet to Infrared

实时动态全息显示


+c
I_S
I_R
LN crystal
Record
I
II
III
IV
V
VI
VII
VIII
IX
Display
+c
I_R
LN crystal
CCD


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光电进展
Vol. 5 No. 1 2022
www.oejournal.org/oea
PIIJ
Institute of Optics and Electronics, Chinese Academy of Sciences

“三维海量光子存储器”



- 存储写入时间：<1ms秒
- 图像分辨率：>400线对/毫米
- 存储量: ~30Gbits/cm^3
- 读取速率: 1MPS
- 固定时间: >20年（实验拟合）

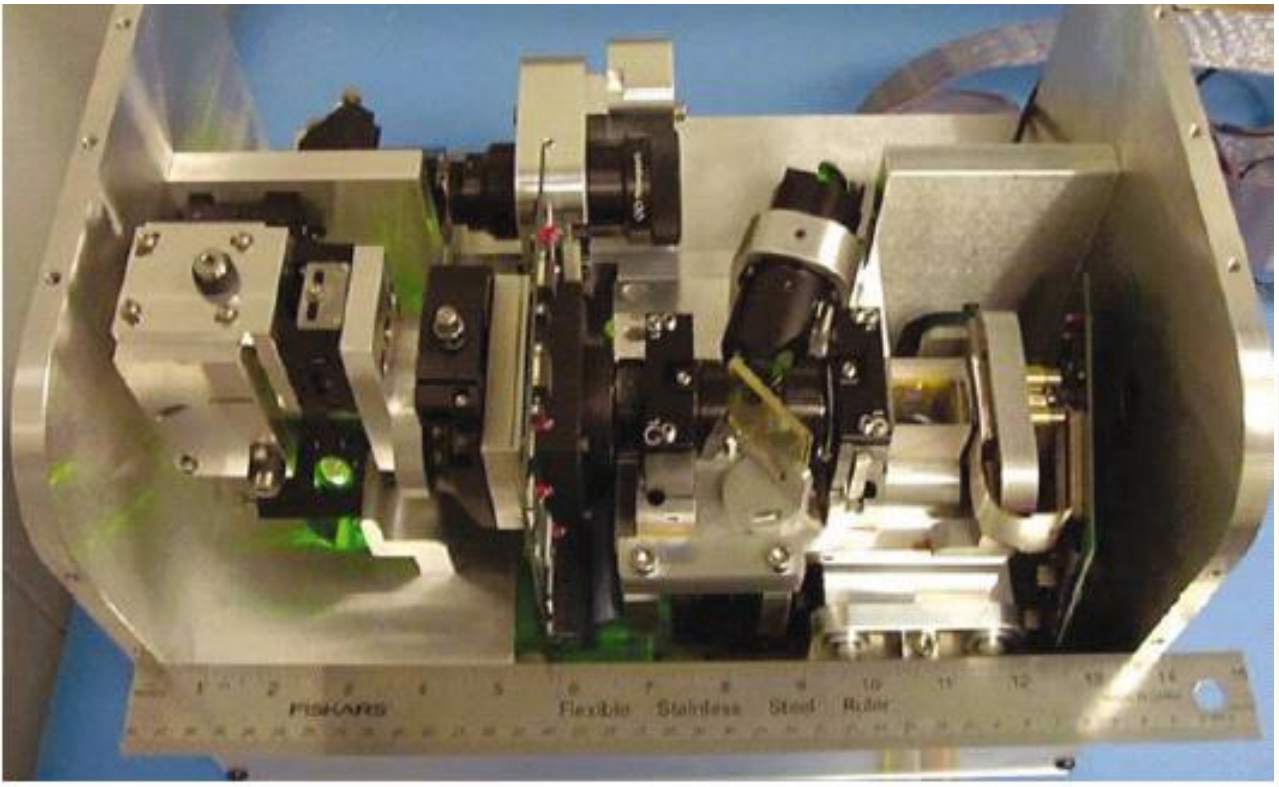
THE FUTURE OF HOLOGRAPHIC STORAGE

Applications of holographic storage have been developed.

- Neural network
- Optical correlator
- Holographic storage for optical communication
- Fiber holographic storage
- Anti-Counterfeit Security



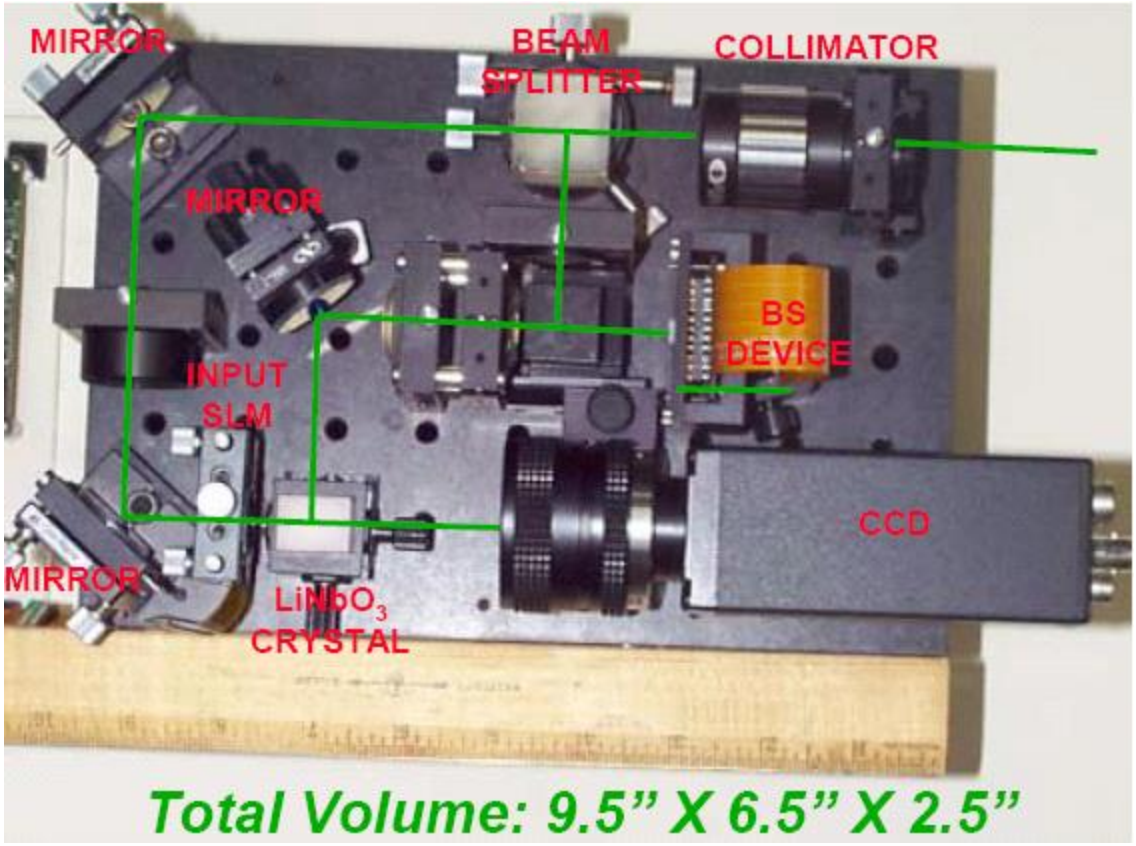
Return

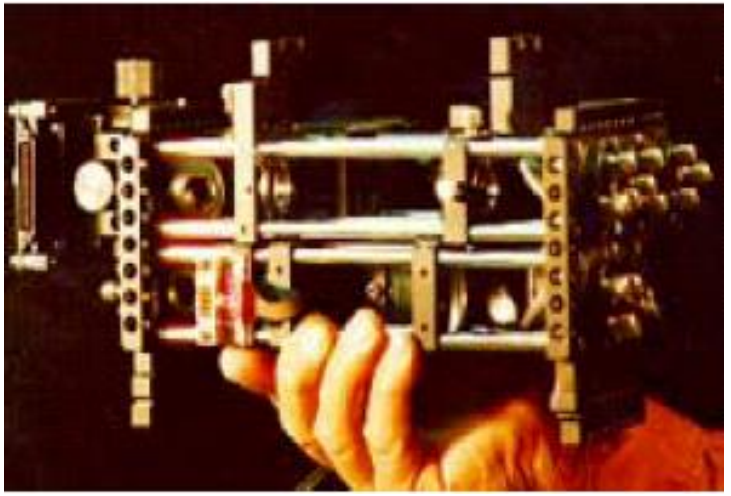


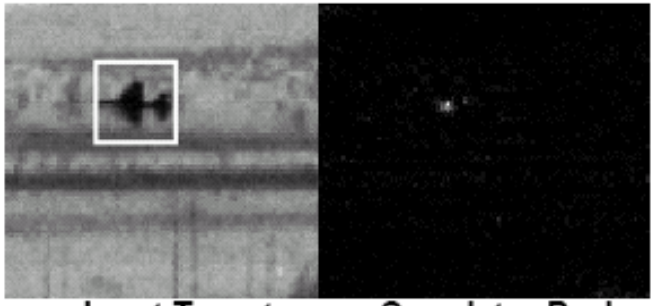
One for posterity: holographic recording systems such as this one could be useful to movie archivers.



Dmitri Psaltis' robot uses a searchable memory of holographic data to help navigate its way home.


MIRROR
BEAM SPLITTER
COLLIMATOR
MIRROR
BS DEVICE
INPUT SLM
CCD
MIRROR
LiNbO3 CRYSTAL
Total Volume: 9.5” X 6.5” X 2.5”

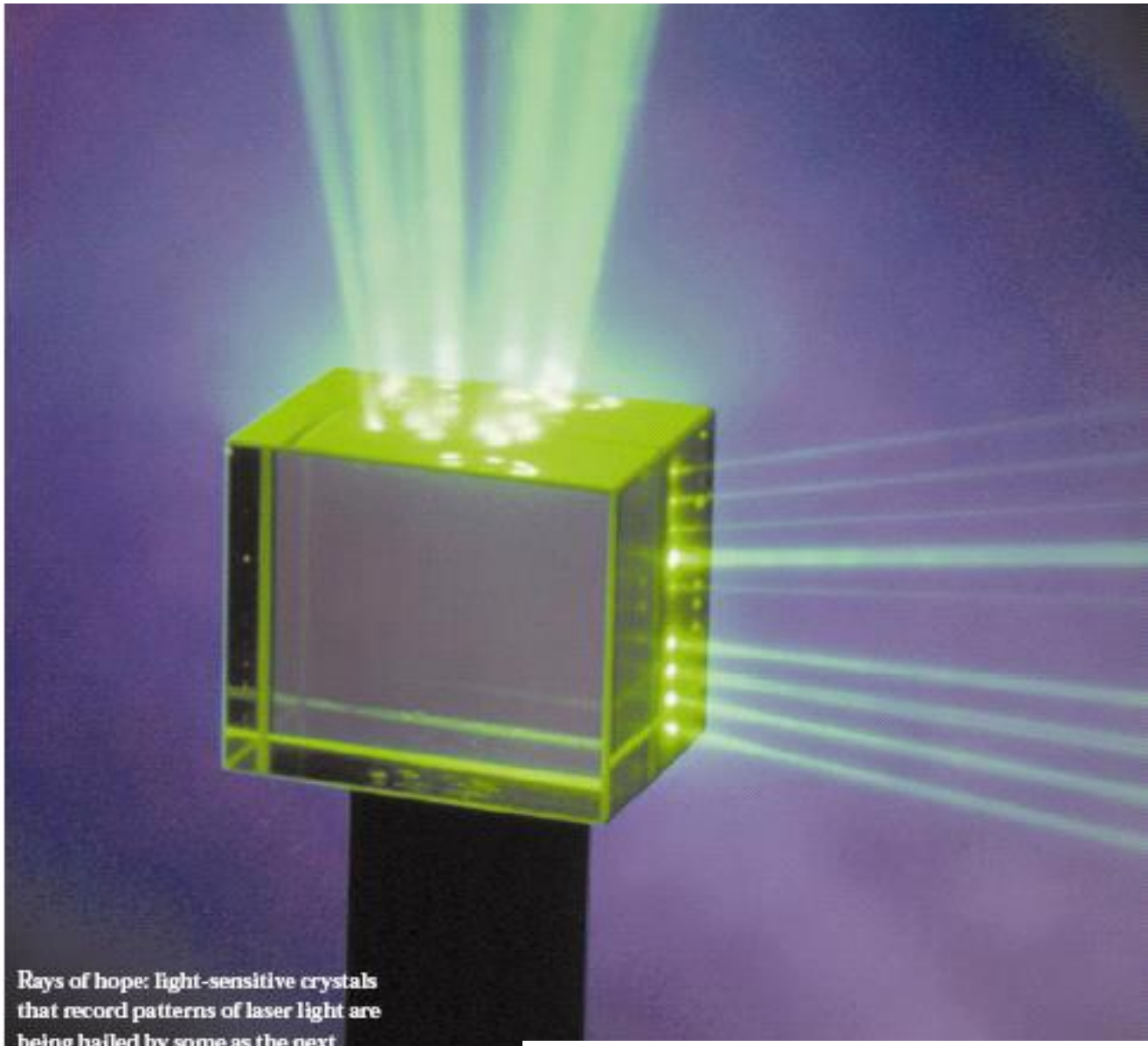



Input Target
Correlator Peak

news feature

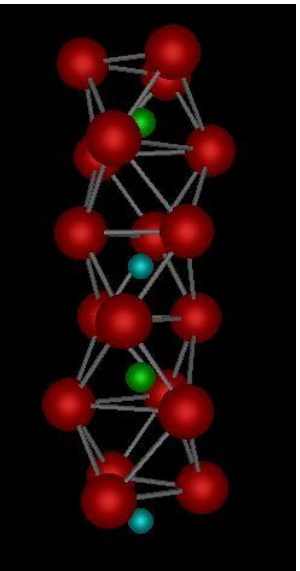
The light fantastic

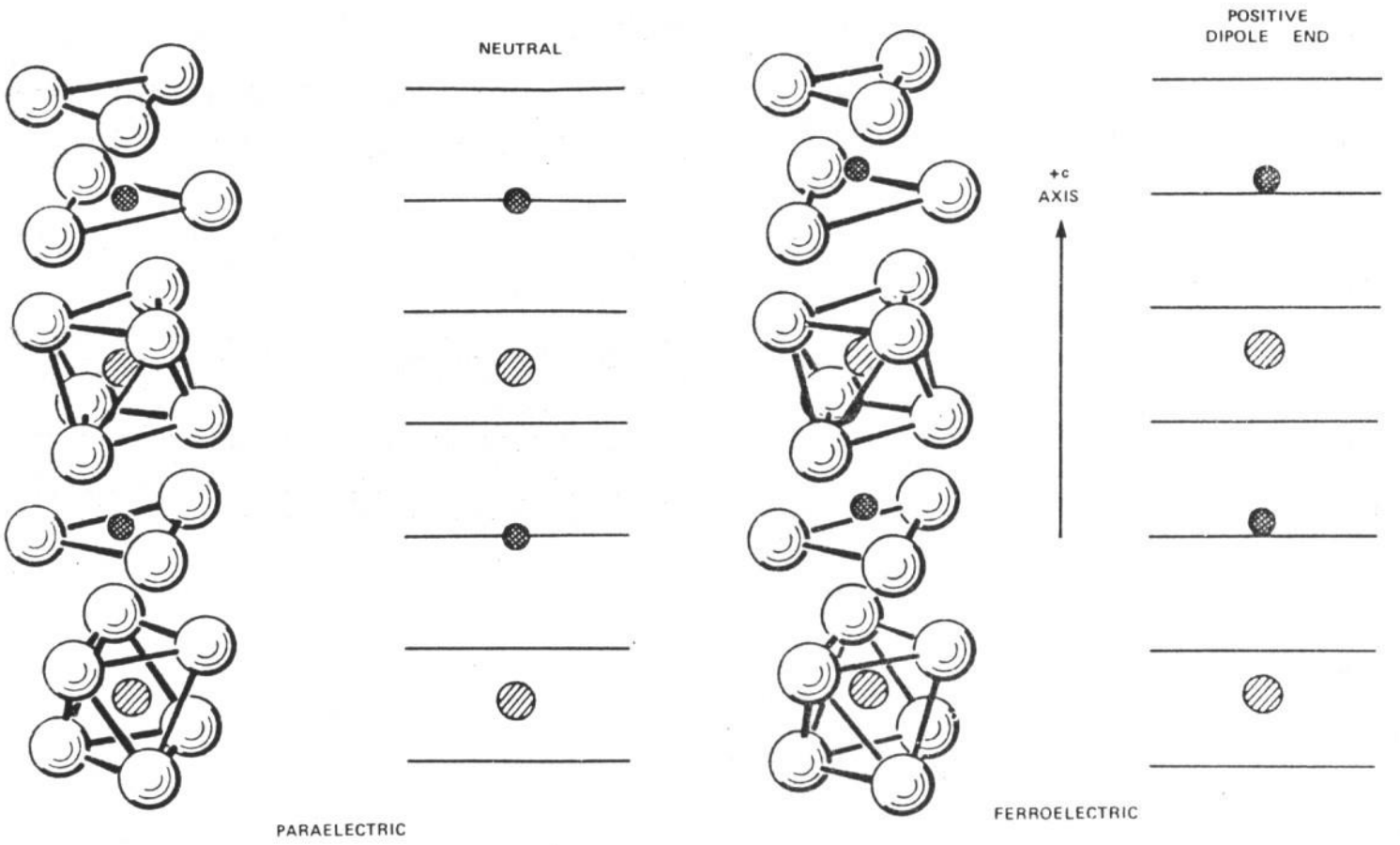
A decade ago, holographic systems promised to revolutionize data storage. The early hype may have evaporated, but the technology quietly progressed, and working devices are now on the market. Mark Haw reports.



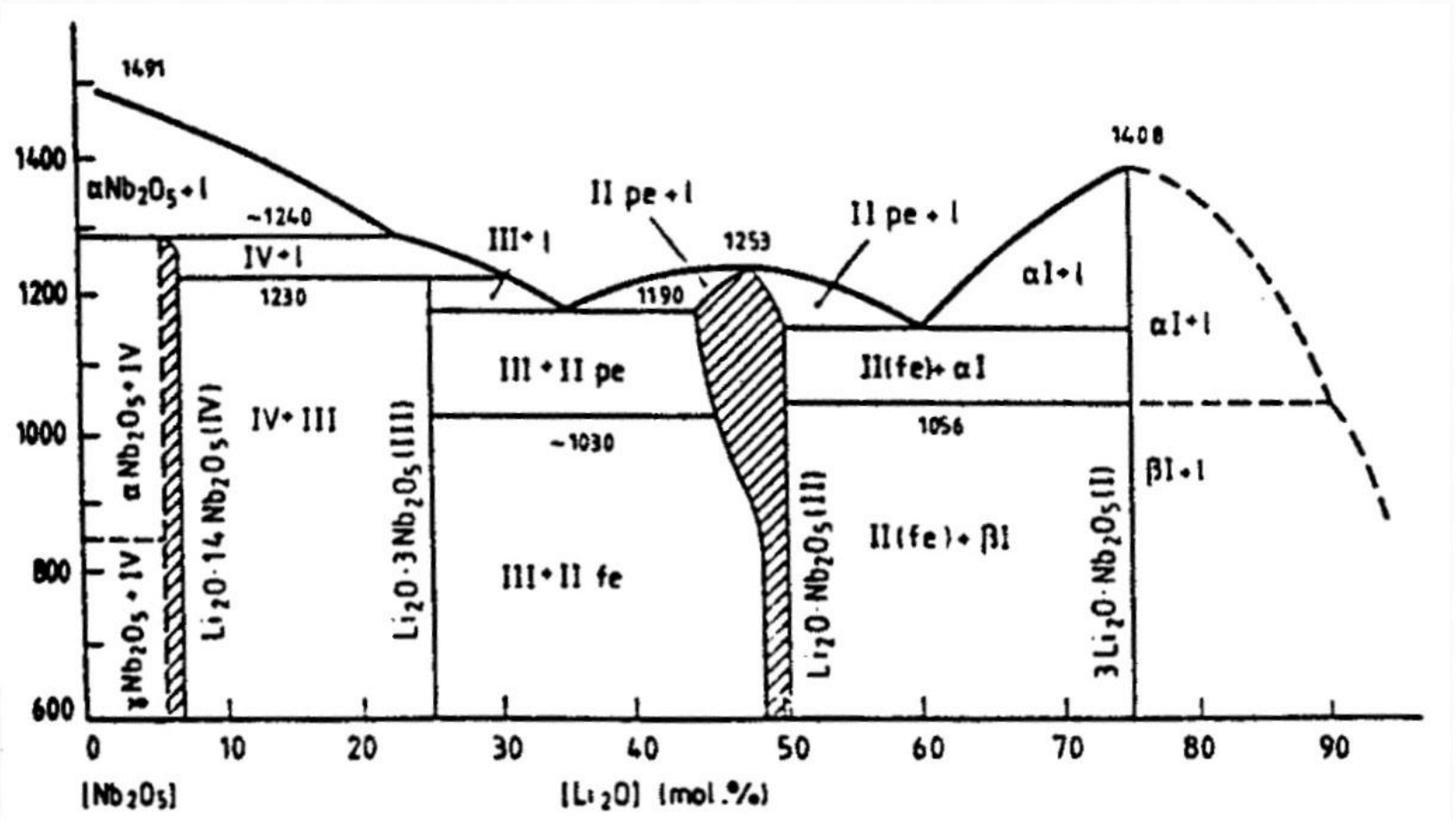
Rays of hope: light-sensitive crystals that record patterns of laser light are being hailed by some as the next generation of data-storage media.

二、基本习性

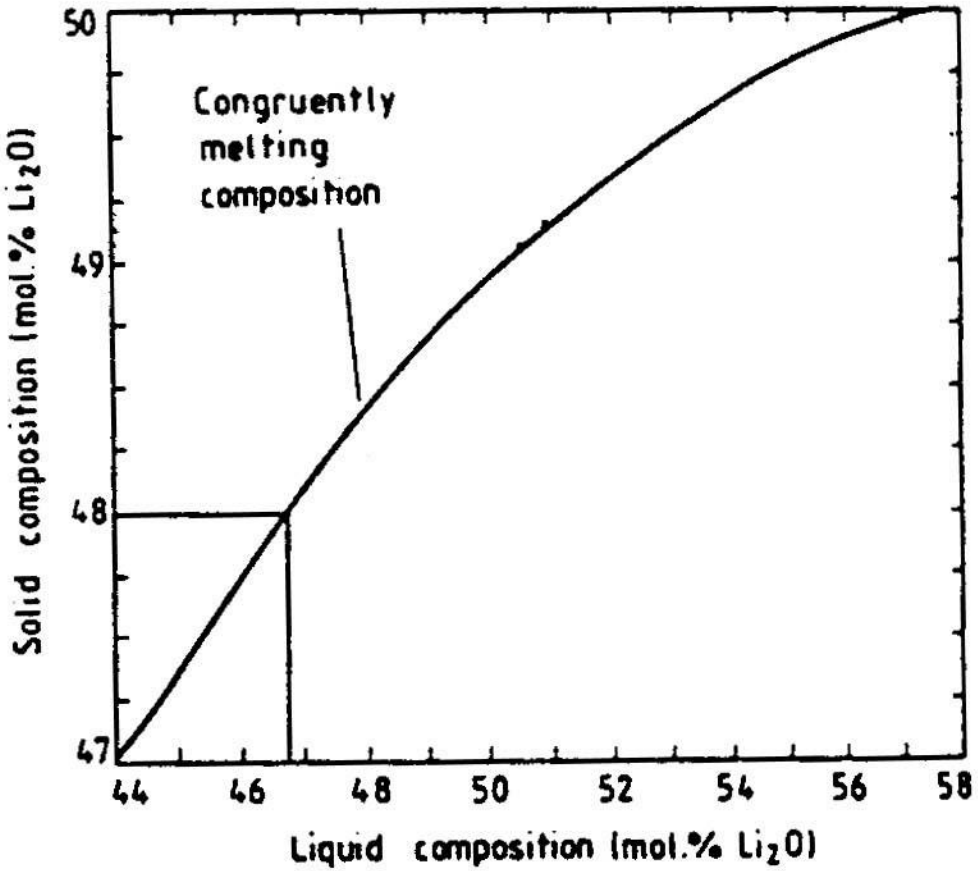



NEUTRAL
POSITIVE
DIPOLE END
+c
AXIS
PARAELECTRIC
FERROELECTRIC

Li_2O - Nb_2O_5系相图



$LiNbO_3$晶体的同成分点



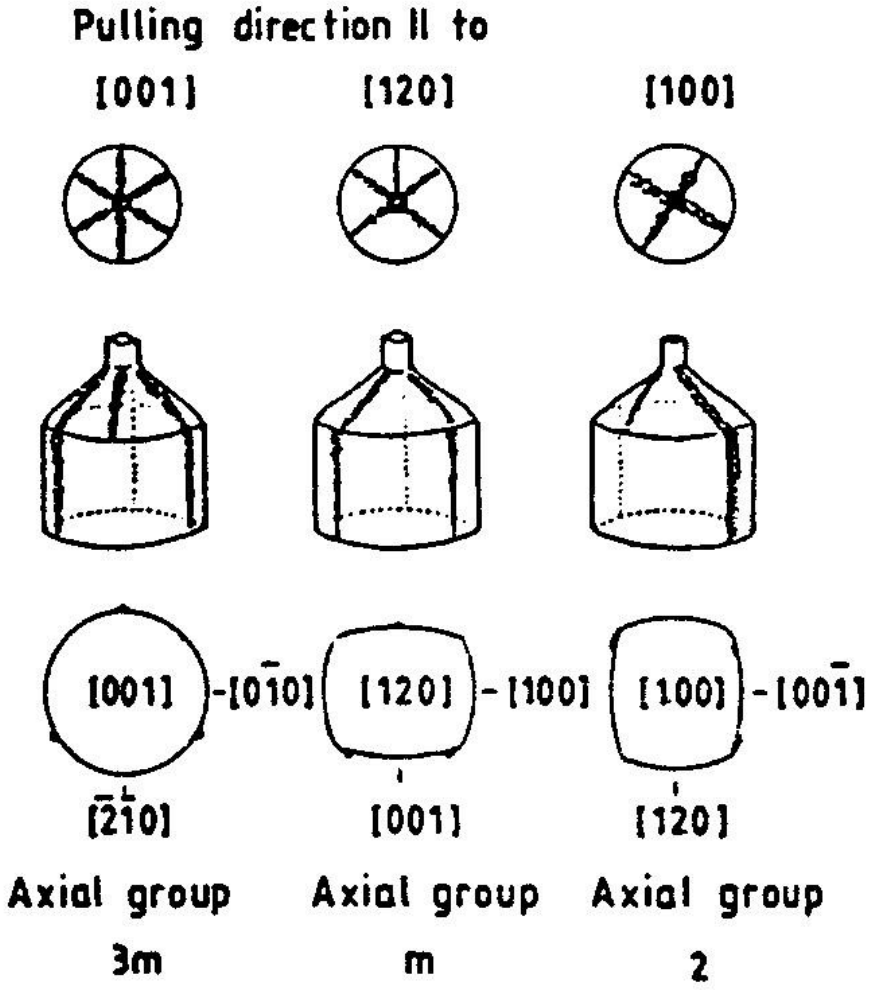
同成分的优势

- 减小了生长晶体的难度，提高了晶体的光学质量。



南开大学
2002年

铌酸锂晶体的生长习性


Pulling direction ll to
[001]
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Axial group
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2

铌酸锂晶体的生长

- 原料的合成，烧料

$$Li_2CO_3 + Nb_2O_5 \rightarrow 2\ LiNbO_3 + CO_2\uparrow$$

- 引晶、拉脖、放肩、等径、收尾
- 高温退火
- 单畴化

单晶生长炉

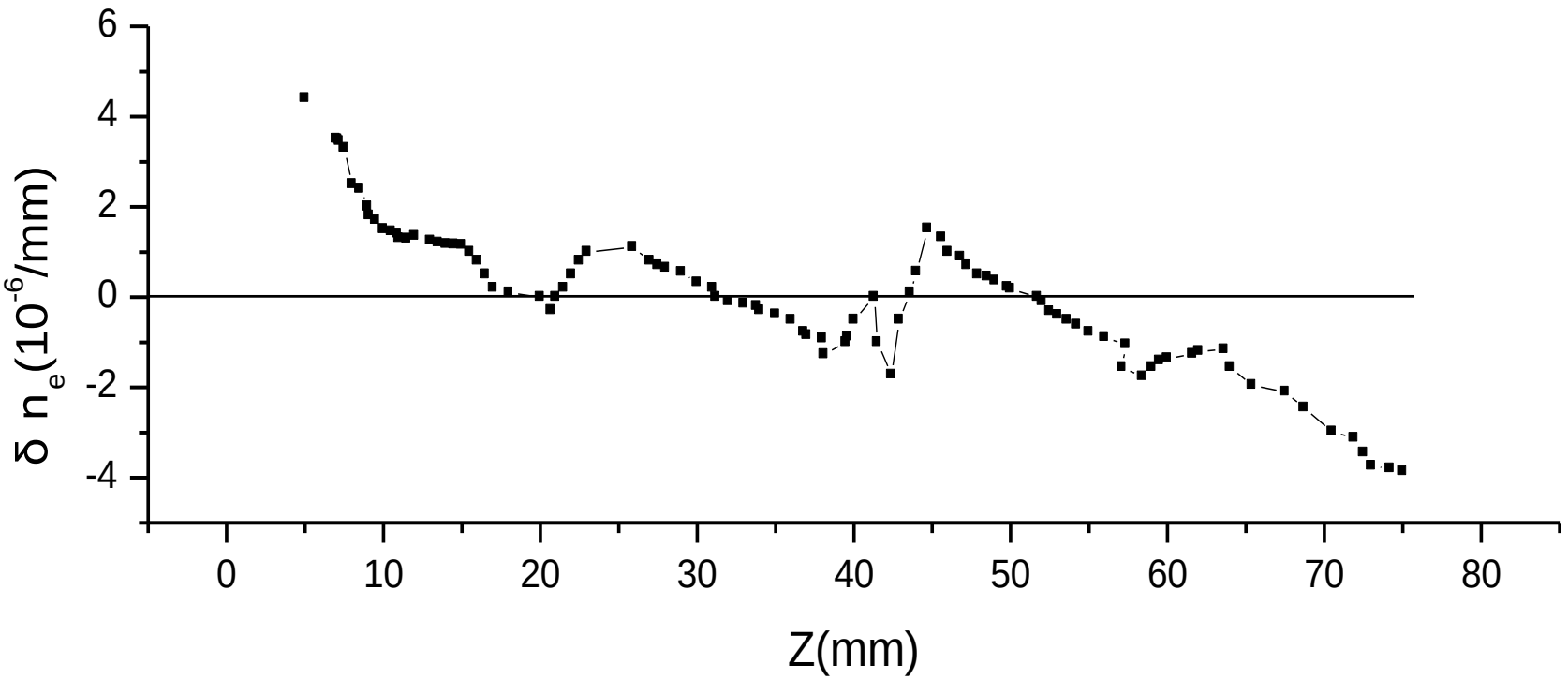




大尺寸光学级铌酸锂晶体



Optical homogeneity



$$\delta\Delta n \approx 1.0\times10^{-5}/\mathrm{cm}$$

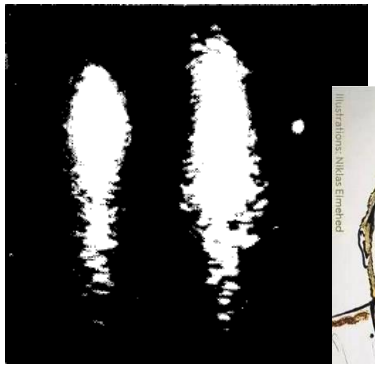
三、掺杂工程

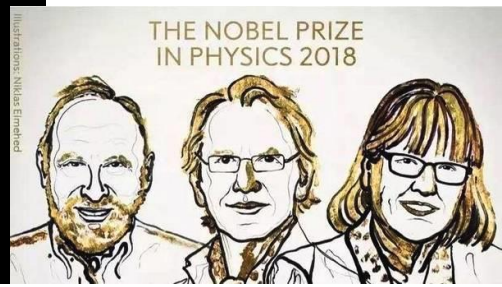
- 抗光损伤掺杂
  - Mg, Zn, In, Sc, Hf, Zr, Sn
- 光折变掺杂
  - Fe, Cu, Mn, Ce, Ni
- 光波导
  - Ti, H^+
- 发光
  - Er^{3+}, Nd^{3+}, Cr^{3+}, Y^{3+}



Optical damage

- Light-induced optical damage, now also named as photorefraction, was discovered in $LiNbO_3$ and $LiTaO_3$ crystals.





Photorefraction (PR)

- ✓ Can be used in
  - holographic storage,
  - information processing,
  - light control of light.
- ✘ low response speed, volatility.

Optical damage

- ✘ Hinders the applications:
  - frequency doublers,
  - optical parametric oscillators,
  - Q-switches,
  - optical waveguides.

A. Ashkin, et al., Appl. Phys. Lett. 9, 72(1966)

“中国之星”

- 1980年，南开大学与西南技术物理研究所合作发现，掺镁量大于4.6 mol%（阈值）时，铌酸锂晶体的抗光折变能力可以提高两个量级以上。
- 1984年在SPIE一次分会上专门对高掺镁$LiNbO_3$晶体的前景进行了讨论；会上，该晶体被誉为“Star of China”。
- 这一原创性的发现引发了国际上的大量研究工作，成为该领域的开篇必引之作，据不完全统计，引用次数已达几百次。

Resist optical damage

- 1980, Mg^{2+} ions, LN:Mg;
  "**Star of China**"
- It promotes the practical applications of LN in nonlinear optics at high light intensities.
- 1990, Zn^{2+} ions, LN:Zn;
- 1992, Sc^{3+} ions, LN:Sc;
- 1995, In^{3+} ions, LN:In.

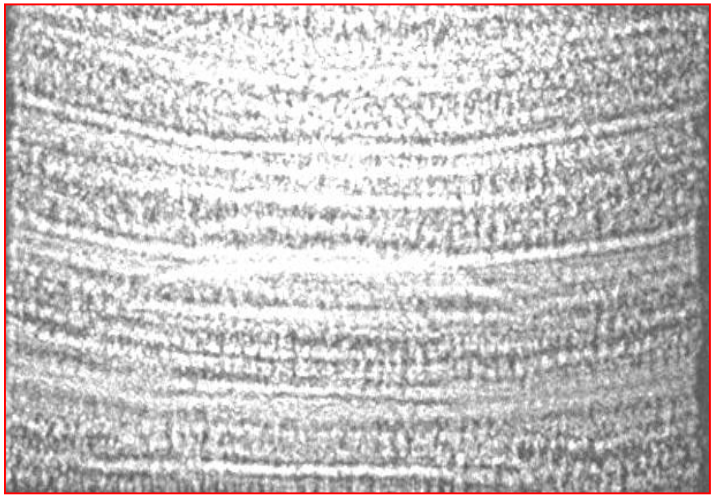


G. Zhong et al., *J. Opt. Soc. Am.* **70**, 631 (1980).
T. R. Volk et al., *Opt. Lett.* **15**, 996 (1990).
J. K. Yamamoto et al., *Appl. Phys. Lett.* **61**, 2156 (1992).
Y. Kong et al., *Appl. Phys. Lett.* **63**, 280 (1995).

The problems of highly doped LN

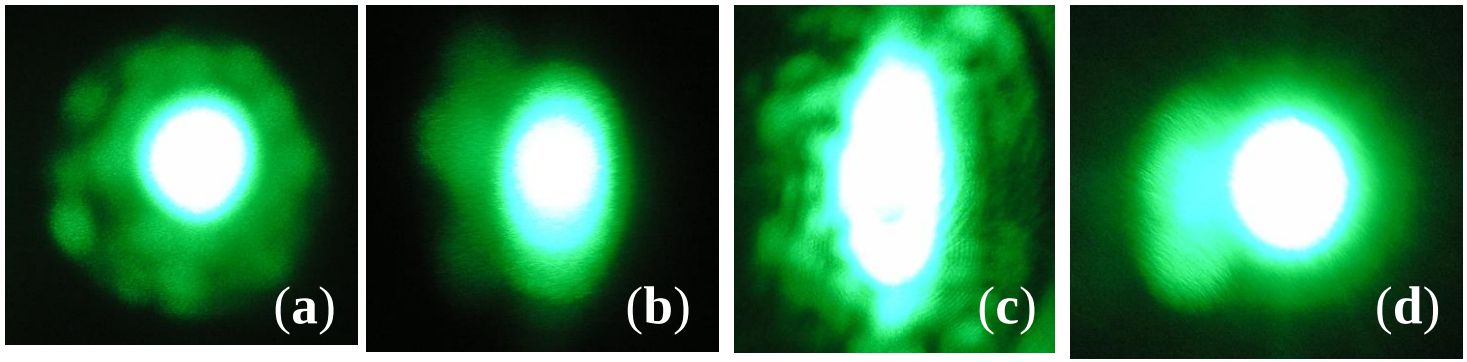
- It is difficult to grow high optical quality crystals.
  - Large amounts of doping concentrations;
    (such as usually 5 mol% Mg for CLN)
  - Distribution coefficient far from 1.0;
    (such as 1.2 for Mg)

- Some properties are still not satisfied:
  - Resistance not high enough,
  - Enhanced ultraviolet photorefraction (UVPR).



ZrO_2 doped $LiNbO_3$ (LN:Zr)

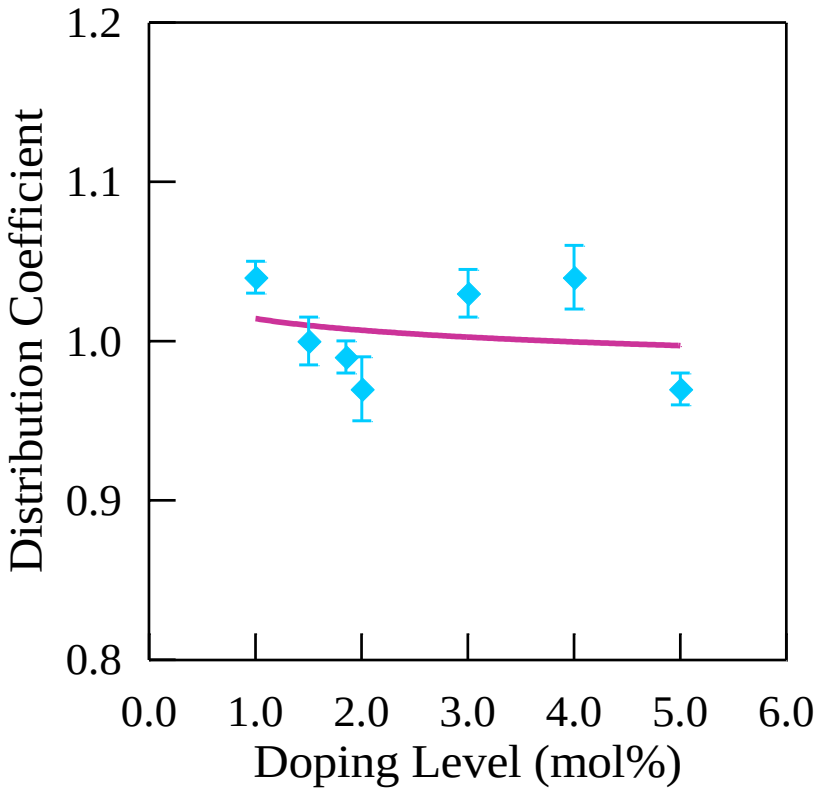
- As the doping concentration reaches 2.0 mol% ZrO_2, LN:Zr crystals can withstand a light intensity as high as 2.0×10^7 W/cm^2.
- At the same experimental conditions, the light intensity that 6.5 mol% Mg doped LN (LN:$Mg_{6.5}$) can withstand is about 5.0×10^5 W/cm^2.



(**a**), (**b**) and (**c**) LN:$Zr_{1.7}$; (**d**) LN: Zr_2. The light intensity for (**a**) is 1.3×10^3 W/cm^2, (**b**) 1.3×10^4 W/cm^2, (**c**) and (**d**) 2.0×10^7 W/cm^2.

Y. Kong et al., Appl. Phys. Lett. 91, 081908 (2007).

The distribution coefficient of Zr



- The maximum value is 1.04 and the minimum value is 0.97.
- Therefore, the distribution coefficient of Zr is much closer to one than that of Mg.

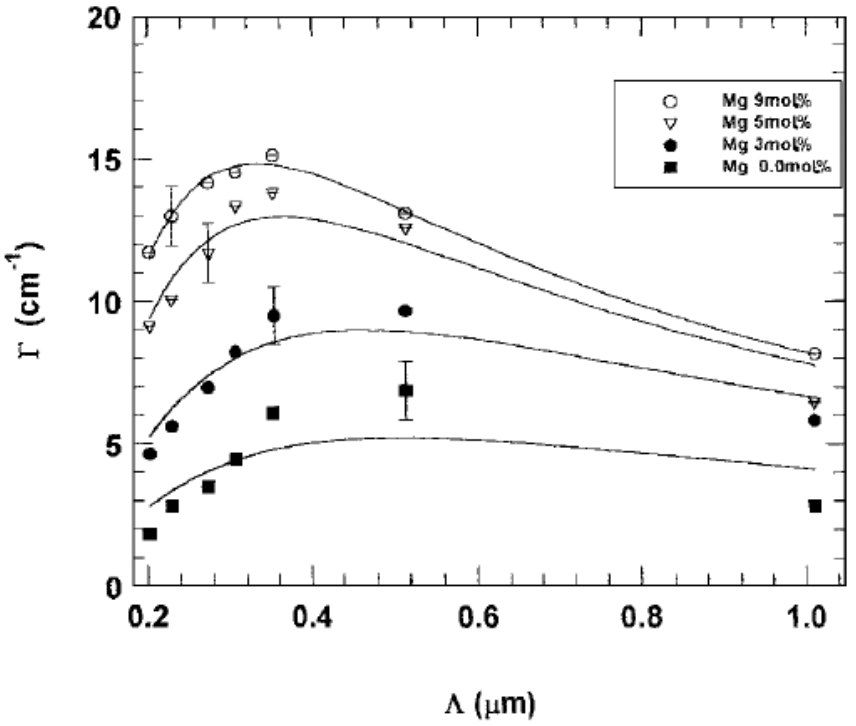
Ultraviolet photorefraction (UVPR)

Enhancement of UVPR in LN:Mg

Table 1. UV Photorefractive Characteristics of Nominally Pure and Mg-Doped $LiNbO_3$ Crystals[a]

Characteristic	MgO-Dopant Concentration (mol.%) 0.0	3.0	5.0	9.0
Absorption coefficient (cm^{-1})	10.2	5.1	2.2	1.2
Specific photoconductivity σ_{ph}/I ($\times 10^{-12}$ cm V^{-2})	0.13	0.25	1.8	2.9
Two-wave mixing gain Γ (cm^{-1})	6.8	10.0	13.8	15.1
Diffraction coefficient η (%)	7.2	13.0	24.6	30.3
Light-induced refractive-index change Δn ($\times 10^{-5}$)	1.0	1.3	1.9	2.1
Photorefractive response time t (s) $(1/e)$	18.4	8.4	1.15	0.6
Photorefractive sensitivity $S = d\Delta n/dIt$ ($\times 10^{-5}$ cm^2/J)	2.6	4.6	15.3	21.6

[a]Experimental conditions, $2\theta \approx 70°$, total light intensity $I =$ 0.2 W/cm, wavelength $\lambda = 351$ nm, and extraordinary polarized writing light.



J. Xu, et al., Opt. Lett. 25, 129(2000)

The resistance of LN:Zr to UVPR

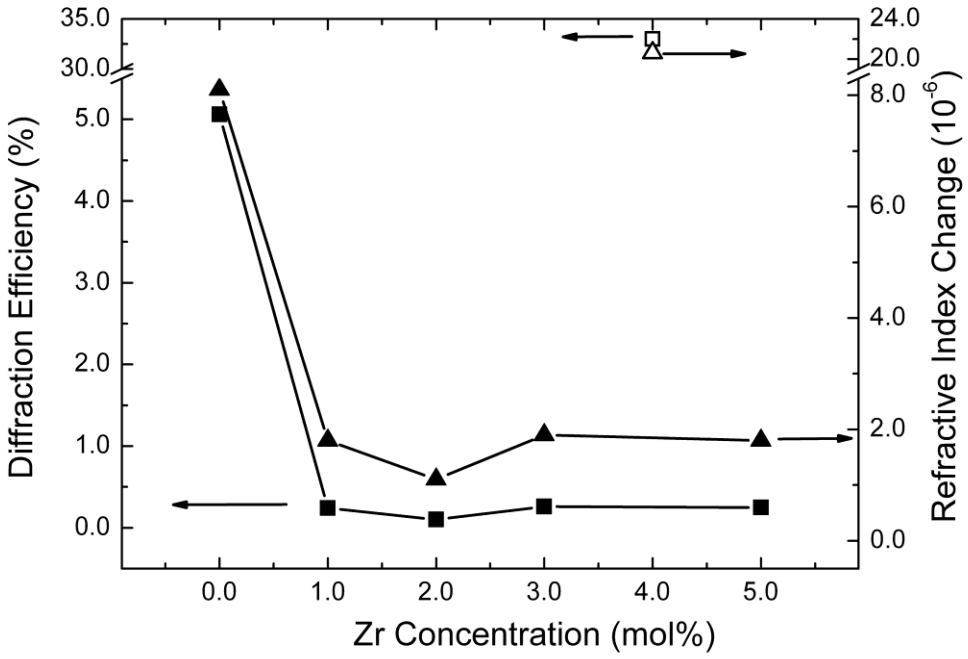


Fig.1 The dependence of UV photorefractive diffraction efficiency and saturated refractive index change of LN:Zr on the doping concentration of Zr.

- The open symbols show the data for LN:Mg5.

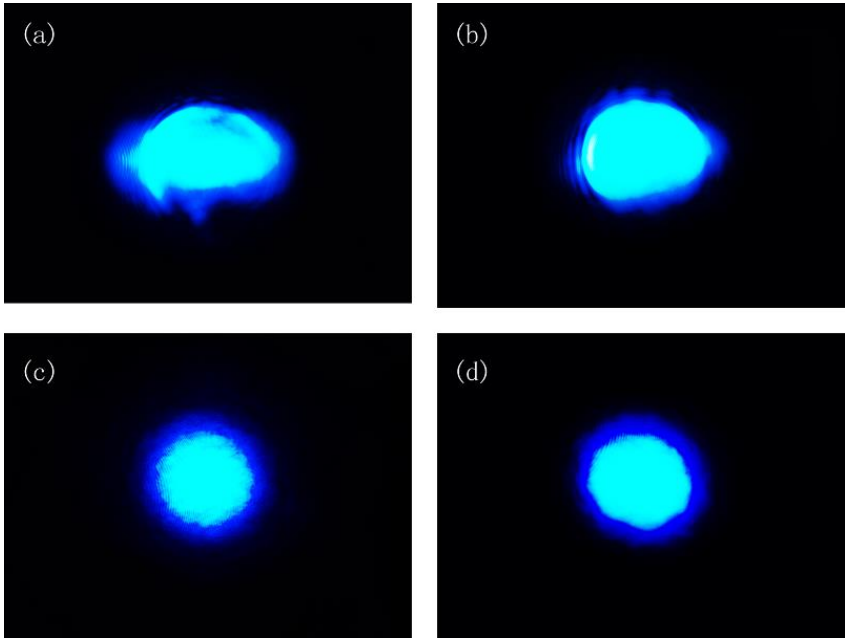


Fig.2 Beam distortion of the transmitted UV light passing through $LiNbO_3$ crystals (wavelength 351 nm, intensity 1.6×10^5 W/cm^2). (a) PLN; (b) LN:Zr1; (c) LN:Zr2; (d) LN:Zr5.

F. Liu, et al., Opt. Lett. 35, 10 (2010)

nature photonics

field electron microscopy', may have many applications for imaging electromagnetic fields in all fields of photonics, but seems especially appealing to those interested in nanostructures and plasmonics.

QUANTUM DOTS

Independent tuning

Appl. Phys. Lett. **95,** 243107 (2009)

Self-assembled quantum dots (QDs) embedded in an optical nanocavity are potentially useful for quantum information processing. It is difficult, however, to couple two QDs to a single nanocavity mode in a cavity quantum electrodynamical system, owing to the linewidth mismatch between the two closely spaced QDs and the cavity modes. Now, Hyochul Kim and co-workers from the USA and the Netherlands have proposed a simple technique that allows the frequencies of individual QDs to be tuned. Their method relied on the use of GaAs photonic crystal membrane structures containing two embedded $In_{0.4}Ga_{0.6}As$ QD layers. The QDs were located in the three-point defects of the photonic crystal, which had a lattice constant and hole radius of 240 nm and 74 nm, respectively. By applying a d.c. voltage of 1.33 V onto only one of the QD layers, the emission line of that particular QD layer was shifted by over 1 nm, leading to an overlap with the cavity mode. Kim and his colleagues are confident that strong coupling effects such as the generation of tripartite quantum states can be achieved by further optimization of their technique.

OPTICAL MATERIALS

UV resistance

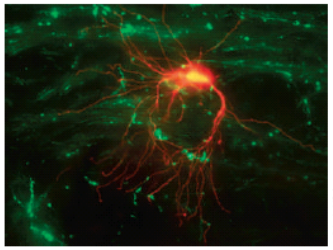
Opt. Lett. **35,** 10–12 (2010)

Although lithium niobate is a useful material for integrated optics and nonlinear photonics, photorefraction — laser-induced optical damage — hinders its applications in the UV region. Now, Fucai Liu and co-workers from Nankai University in China have carried out a systematic investigation into the UV-range damage resistance of lithium niobate doped with zirconium. For comparison, three kinds of lithium niobate were grown by the Czochralski method: Zr-doped, MgO-doped and congruent pure lithium niobate. Two-wave mixing techniques were used to measure both the increase in refractive index when irradiated by 351 nm light and the decrease in refractive index when irradiated by 532 nm light. The researchers found that the Zr dopant was the most efficient for preventing optical damage at UV wavelengths, and a sample with 2.0 mol% Zr dopant was reported to have a damage resistance threshold of 10^5 W cm^{-2}.

NEUROSCIENCE

Light and headaches

Nature Neurosci. **13,** 239–245 (2010)



© 2010 NPG

Many migraine sufferers experience photophobia — an abnormally high sensitivity to light — during intense migraine episodes. Rodrigo Noseda and his colleagues have now identified the neural mechanism responsible for this painful phenomenon. By examining a group of blind migraine sufferers, the researchers determined that those who had lost their optic nerves or both of their eyes did not suffer from photophobia. They then conducted various electrophysiological neuronal tests in rats. Through their experiments, the researchers concluded the neuronal activity of the noci pathway — which causes experienced during — is regulated by retin activation. This ine sig from light-sen ganglion cells that con on the dura-sensitive neurons in the brain, which are responsible for migraine pain.

BIOPHOTONICS

Neural silencing

Nature **463,** 98–102 (2010)

Multicoloured lights are capable of silencing brain activity, according to findings reported by Brian Chow and co-workers at the Massachusetts Institute of Technology in the USA. The researchers introduced light-driven proton pumps — proteins embedded in cell membranes that transport molecules in and out of the cells — to neurons in the brain. These proton pumps respond to illumination with specific colours of light; the two pumps chosen for this experiment — Arch and Mac — were sensitive to yellow–green and blue–green light, respectively. The team showed that irradiating the neurons with either yellow or blue light caused protons to be removed from the cell, which inhibited neural activity. They also discovered that light-driven proton pumps were able to mediate significantly higher photocurrents (approac pA) than light-driven chloride pump ermore, Arch-mediated current wn to recover from yellow light illum within a matter of seconds — much n any of the other proton pumps prev ted. The researchers suggest that their fi could lead to experiments where two or more neural projections from a single site in the brain could independently be silenced during analysis, which could lead to a greater understanding of the underlying causes behind brain activity and more effective treatments for neural disorders.

PLASMA PHYSICS

Collisionless shocks

Phys. Rev. Lett. **103,** 255001 (2009)



Emissions of plasma from superm gamma-ray bursts and relativistic interact with the surrounding int medium through relativistic colli shocks. Despite the importance o shocks and their associated instab large number of astrophysical sit there is currently no single, comp analytical theory to describe ther Karl Krushelnick and his co-wor the UK, Greece and Germany hav achieved the first laboratory dem of such ultrahigh-velocity shocks focusing 350-TW, 1-ps laser pulse energy in the range of 200–350 J edge of a low-density helium gas The resulting electrostatic shock into the surrounding helium gas, three distinct stages. In particula researchers found that the third s shock — known as the blast wave unlike any laser-generated blast w previously demonstrated in the la because thin-shell perturbations to grow without fragmentation o blast front.

OPTICAL MATERIALS

UV resistance

Opt. Lett. **35,** 10–12 (2010)

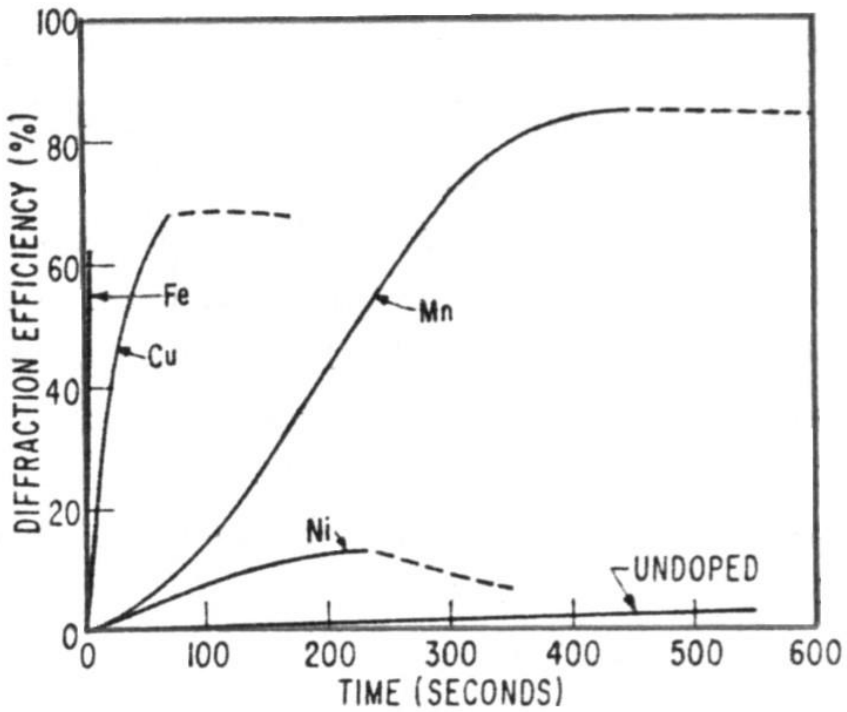
Although lithium niobate is a useful material for integrated optics and nonlinear photonics, photorefraction — laser-induced optical damage — hinders its applications in the UV region. Now, Fucai Liu and co-workers from Nankai University in China have carried out a systematic investigation into the UV-range damage ped with e kinds y the MgO- m niobate. re used refractive n light dex when researchers most amage e with d to have a 10^5 W cm^{-2}.

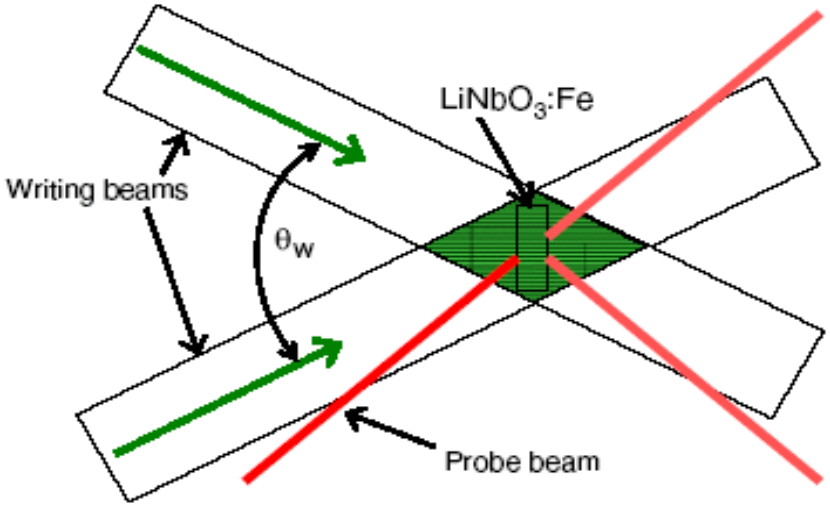
抗紫外光损伤

虽然铌酸锂晶体对于集成光学和非线性光子学是一种有用的材料，但是光折变（光损伤）阻碍了它在紫外光区的应用。来自中国南开大学的刘富才及合作者对于掺锆铌酸锂晶体抗紫外光损伤进行了系统的研究。…

研究者们发现掺锆对于阻止紫外光损伤最有效，其中掺2.0mol%锆晶体的光伤阈值达到了10^5 W/cm^2。

Enhance photorefraction





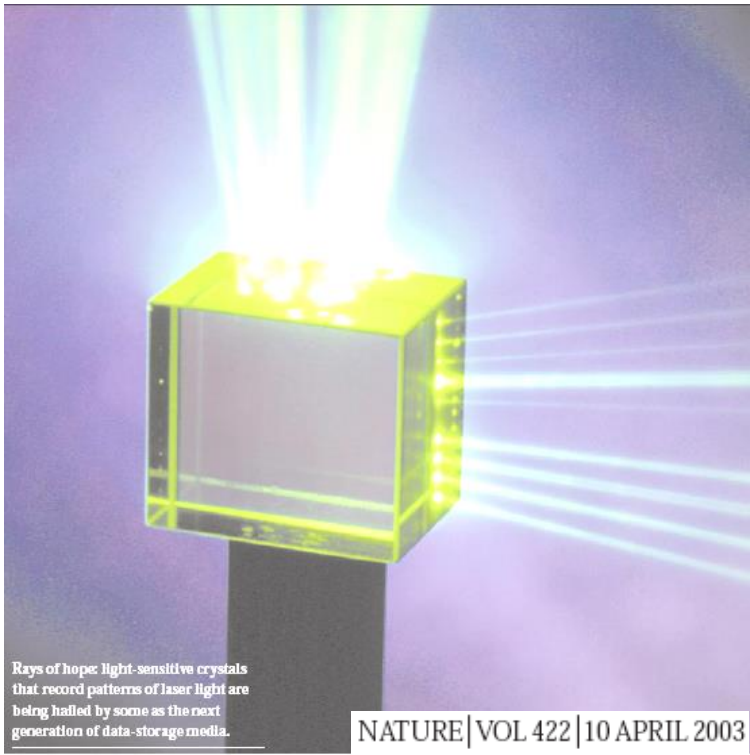
Writing (solid line) and erasing (dash line) curves for several doped lithium niobate crystals. Wavelength 488 nm， light intensity 400 mW/cm^2

W. Phillips et al., RCA Review 33, 94 (1972).

Fe_2O_3 doped $LiNbO_3$ (LN:Fe)

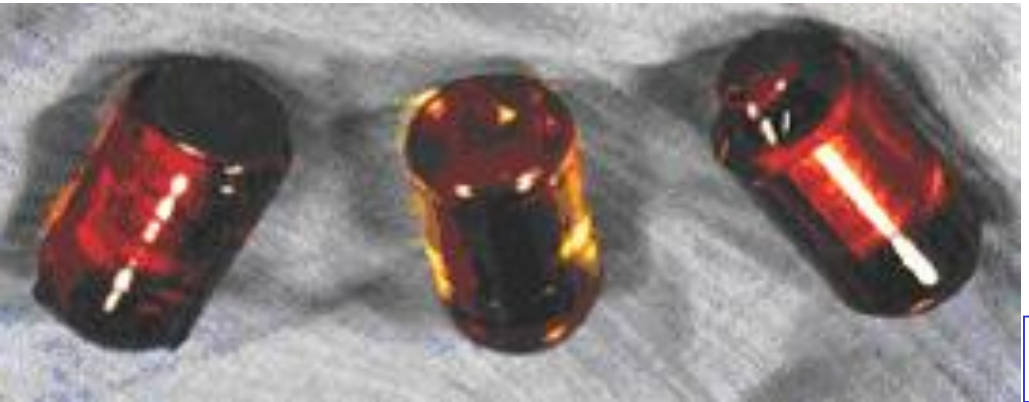
- By now, Fe_2O_3 doped $LiNbO_3$ (LN:Fe) is one of the most excellent candidate materials for optical data storage due to its:
  - ✓ high diffraction efficiency,
  - ✓ high data storage density,
  - ✓ long storage lifetime.

- The problems:
  - low response speed,
  - strong light-induced scattering,
  - volatility.



A solution to increase the response speed

- Co-doping with damage-resistant elements such as Mg, Zn, In and Sc, has been found to be a useful way to increase the response speed and resistance to scattering.
- When the doping concentrations are above the threshold, Fe^{3+} ions and part of Fe^{2+} ions on Li sites will be repelled to Nb sites,
  - ✓ improves the response speed.
  - ✘ apparently decreases the diffraction efficiency.



G. Zhang, *Proc. SPIE* **2529**, 14 (1995).

HfO_2 and Fe_2O_3 co-doped $LiNbO_3$ (LN:Fe,Hf)

Samples	Doping concentrations			Photorefractive properties		
	Fe (wt.%)	Mg (mol%)	Hf (mol%)	η_{sat} (%)	τ_r (s)	S (cm/J)
LN:Fe	0.01			70	160	
LN:Fe:Mg_2	0.01	2		70	60	
LN:Fe:Mg_6	0.01	6		15	15	
LN:Fe:Hf_2	0.03		2	68.0	17.2	3.99
LN:Fe:Hf_4	0.03		4	47.6	12.6	4.36
LN:Fe:Hf_5	0.03		5	55.4	10.7	5.23

S. Li, et al., *Appl. Phys. Lett.* 89, 101126 (2006)

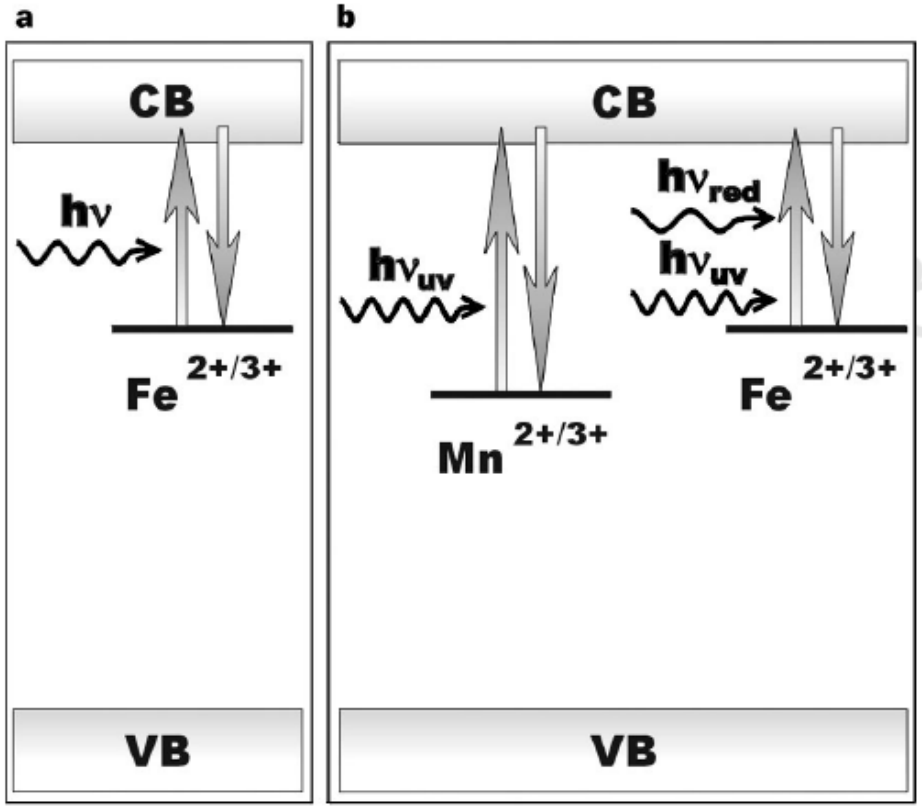
ZrO_2 and Fe_2O_3 co-doped $LiNbO_3$ (LN:Fe,Zr)

Samples	Doping concentrations			Photorefractive properties		
	Fe (wt.%)	Mg (mol%)	Zr (mol%)	η_{sat} (%)	τ_r (s)	S (cm/J)
LN:Fe	0.01			70	160	
LN:Fe,Zr_1	0.03		1	25.5	2.2	13.46
LN:Fe,Zr_2	0.03		2	32.0	1.8	12.87
LN:Fe,Zr_3	0.03		3	32.7	1.8	13.48
LN:Fe,Zr_4	0.03		4	32.5	1.8	13.40
LN:Fe,Zr_5	0.03		5	42.2	2.2	12.61

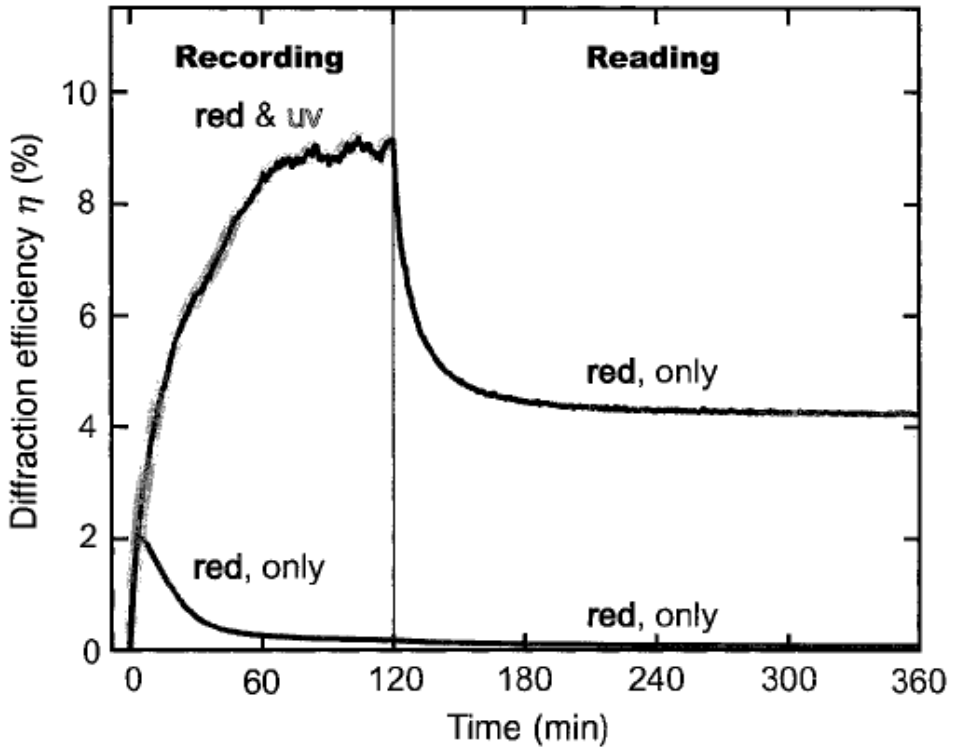
Y. Kong et al., *Appl. Phys. Lett.* 92, 251107 (2008)

Nonvolatile holographic storage

$LiNbO_3$:Fe,Mn

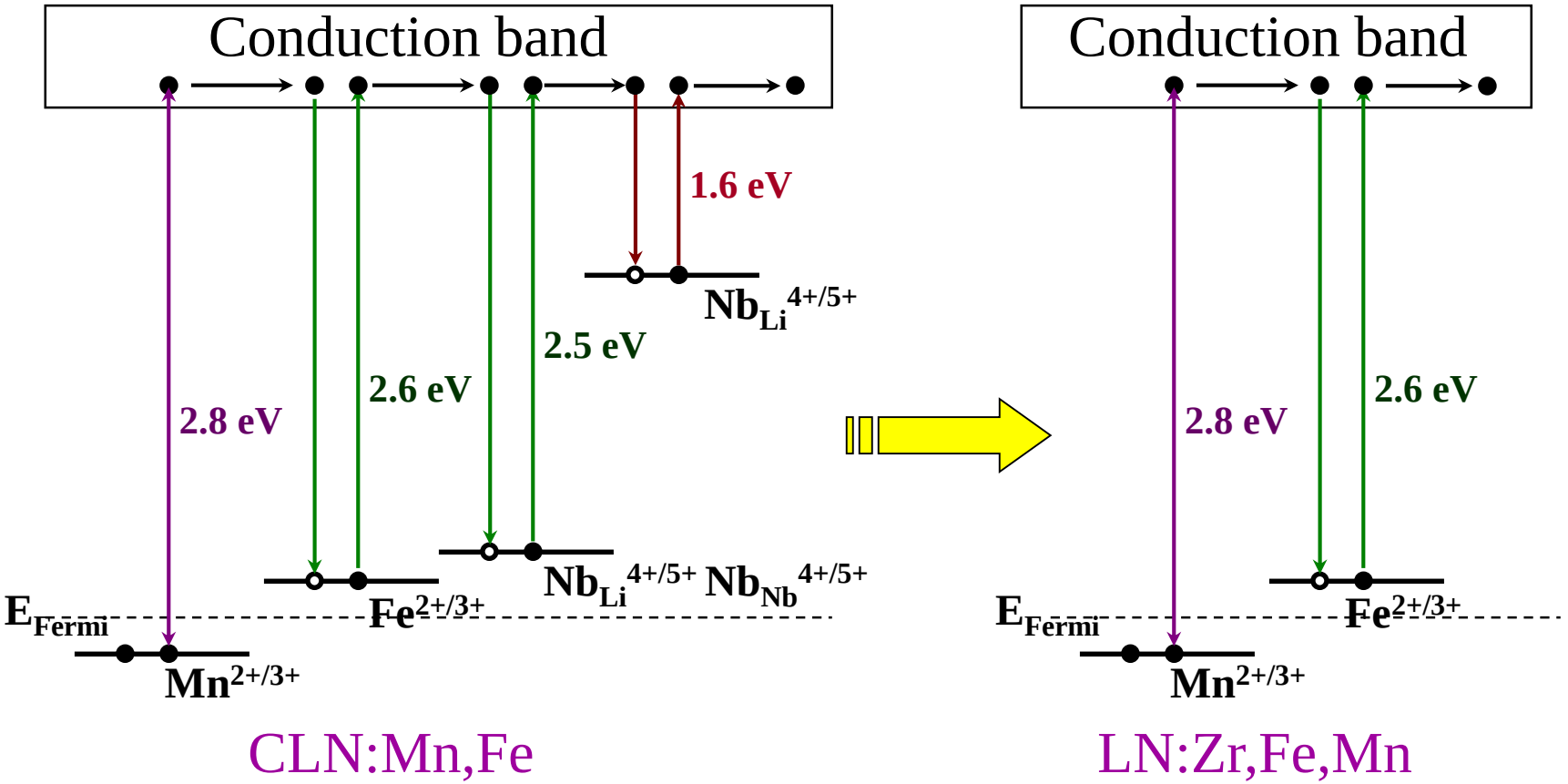


one-center two-center



K. Buse, et al., Nature 393, 665 (1998)

Energy level diagram of $LiNbO_3$

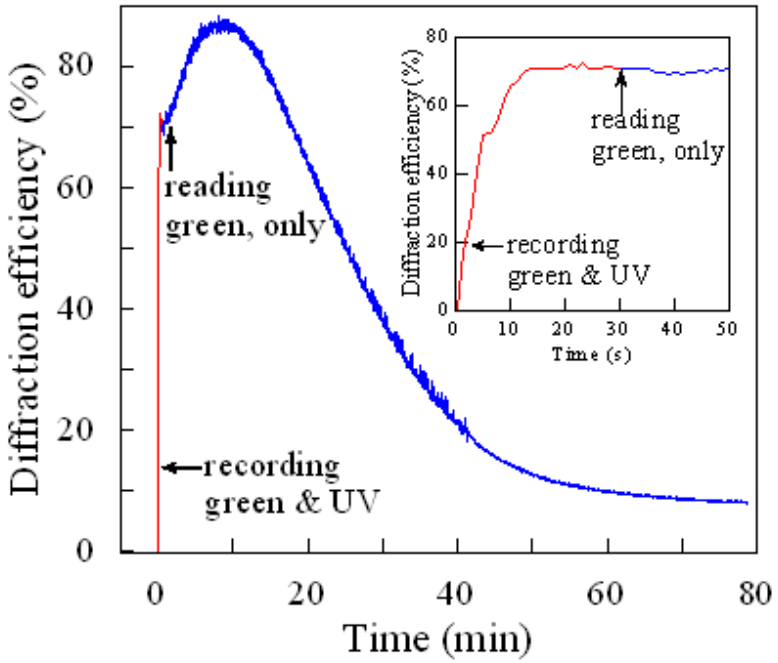


CLN:Mn,Fe

LN:Zr,Fe,Mn

The co-doping of Zr eliminates undesired intrinsic electron traps, which greatly enhances the charge transition speed for nonvolatile holographic storage

$LiNbO_3$:Zr,Fe,Mn



Oxidation time	I_{rec}/I_{sen} (mW/cm^2)	η_s (%)	η_f (%)	S' (cm/J)	τ_r (s)
24h	800/40	54.3	14.9	0.65	2.4
24h	600/40	52.1	14.5	0.88	2.2
24h	400/40	62.0	13.6	1.78	1.2
48h	400/40	57.0	7.8	1.13	2.0
56 20h	400/40	62.5	14.0	2.10	0.88


饶毓泰基础光学奖　刘富才　同志撰写的

荣誉证书

《掺锆铌酸锂晶体的光折变性能研究》

科学论文

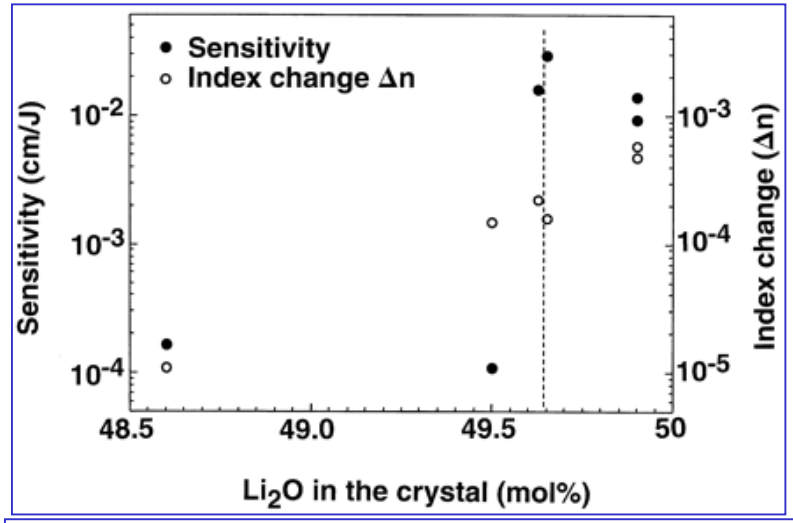
荣获　第七届

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1891—1968

评委会主席

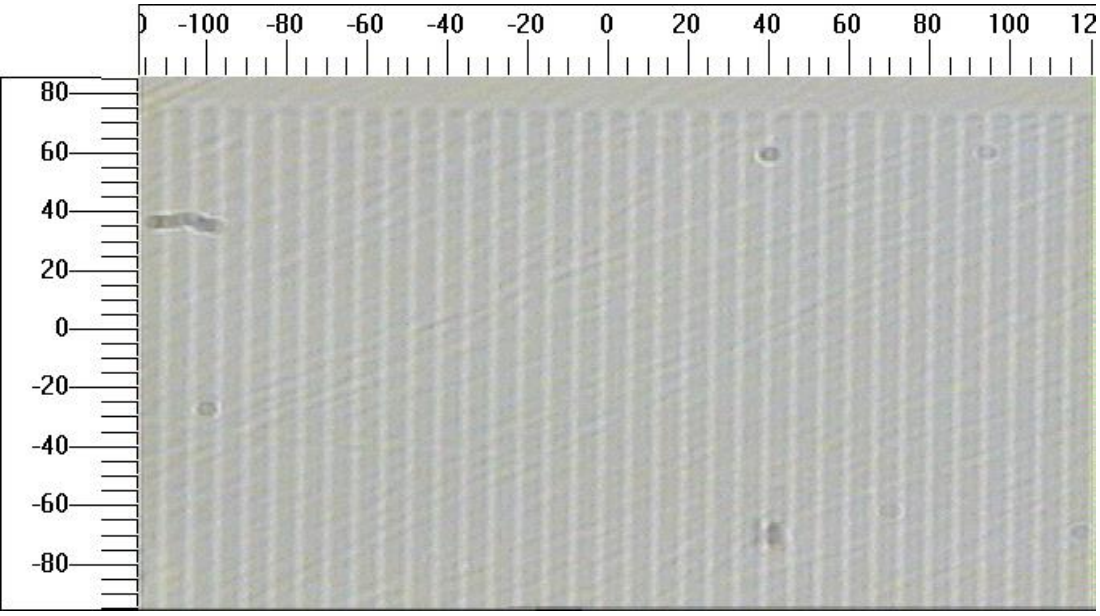
证书序号　2010-8

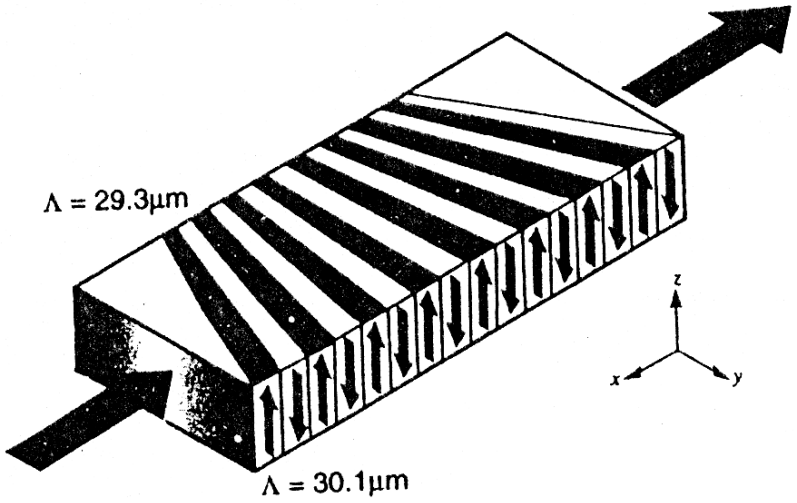


L. Hesselink, et al., Science 282, 1089 (1998)

Y. Kong et al., *Opt. Lett* 34, 3896 (2009)

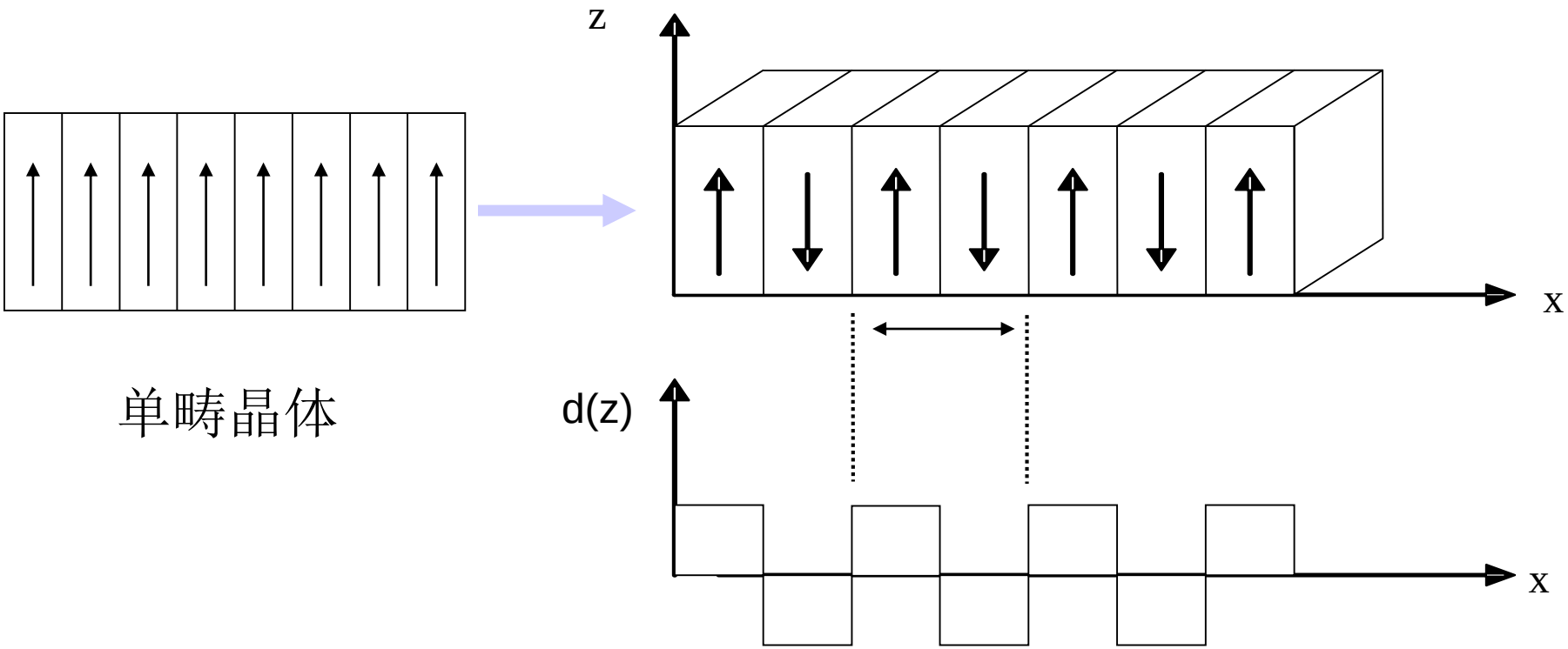
四、畴工程





- 周期极化铌酸锂激光波长转换器

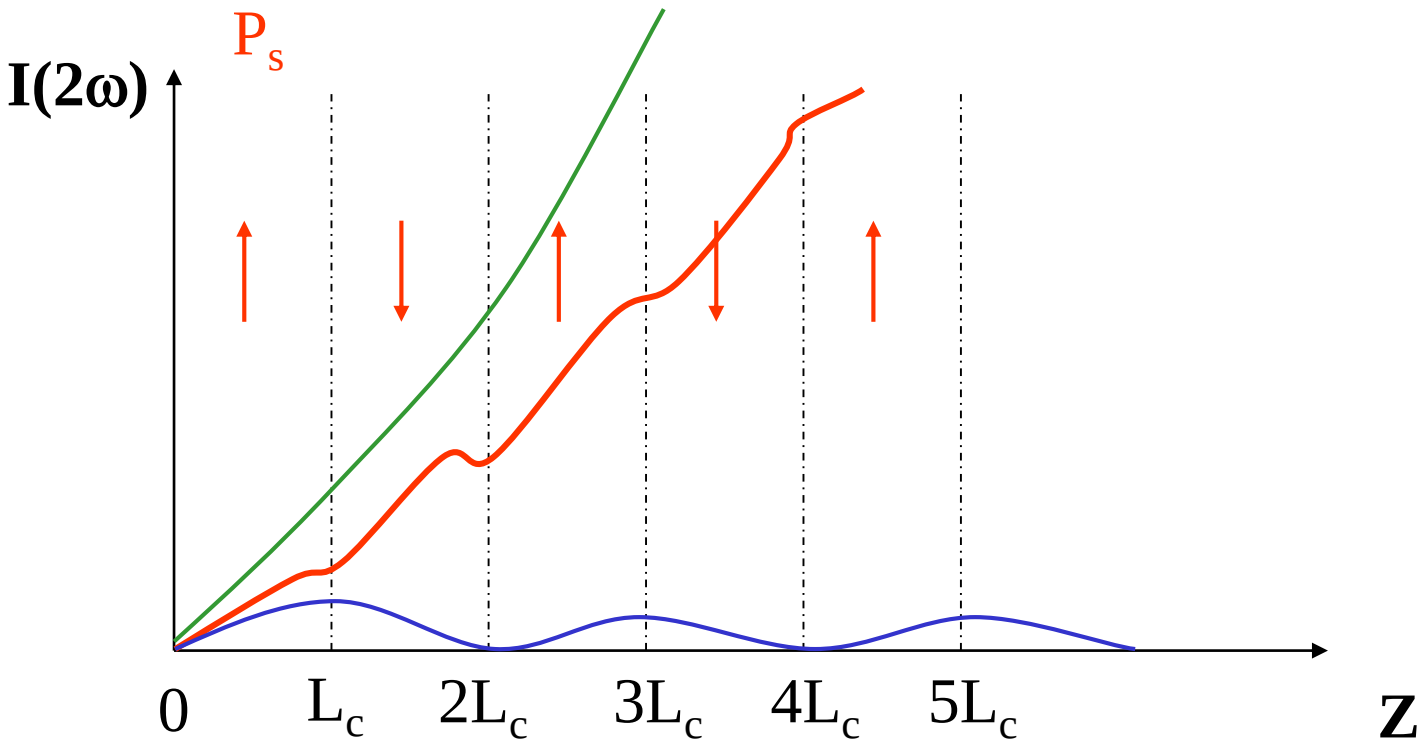
准相位匹配原理



单畴晶体

利用周期极化晶体实现QPM的原理示意图

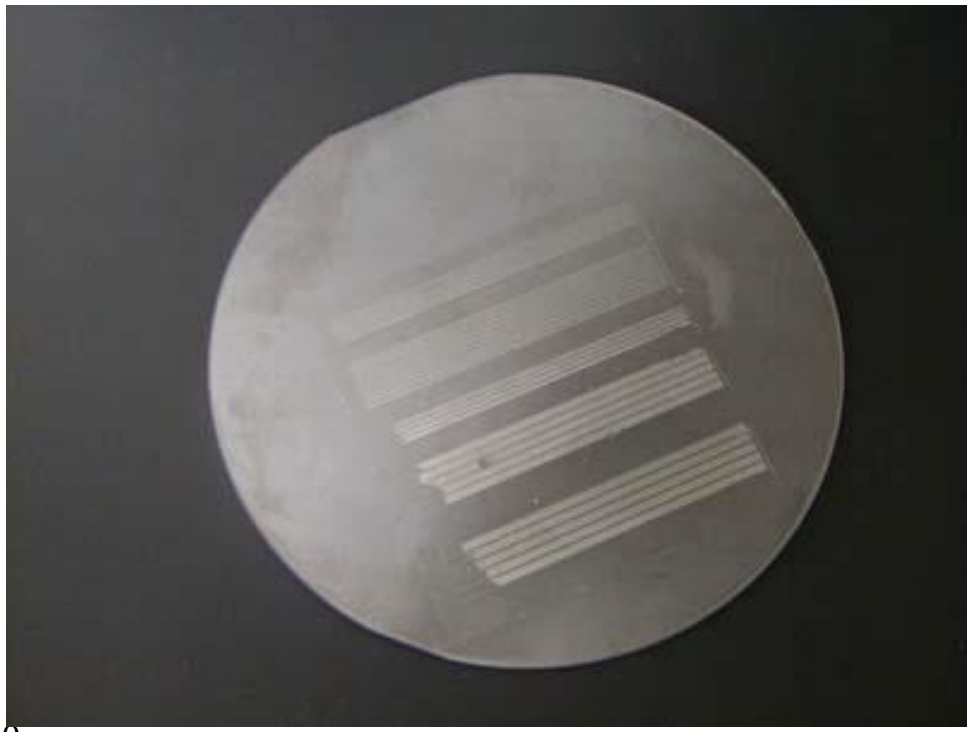
理想PM, QPM和无PM下的倍频光强千随介质长度的变化

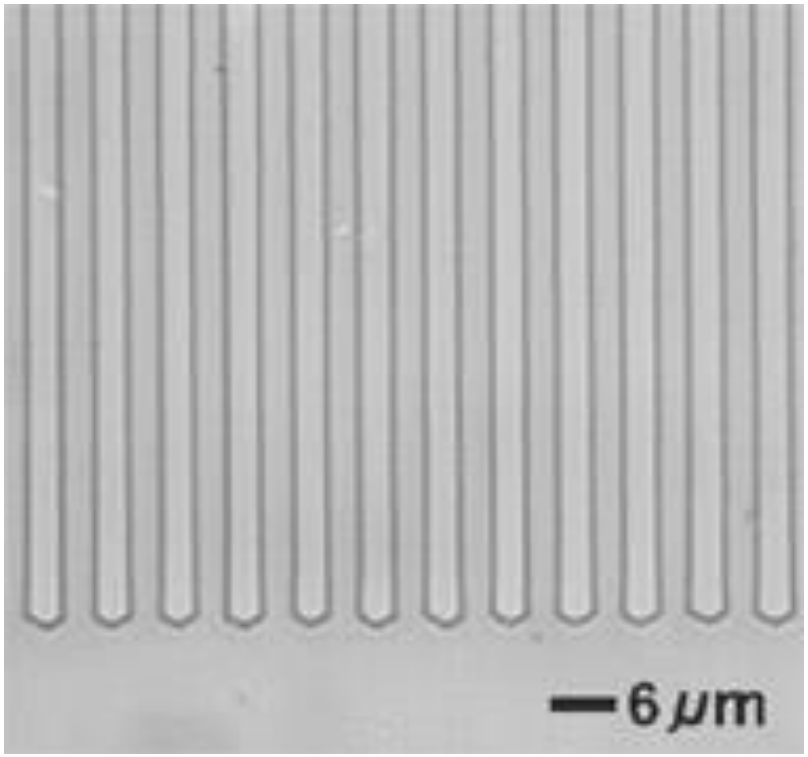


$$L_c = \lambda_0^{\omega} / 4(n^{2\omega} - n^{\omega})$$

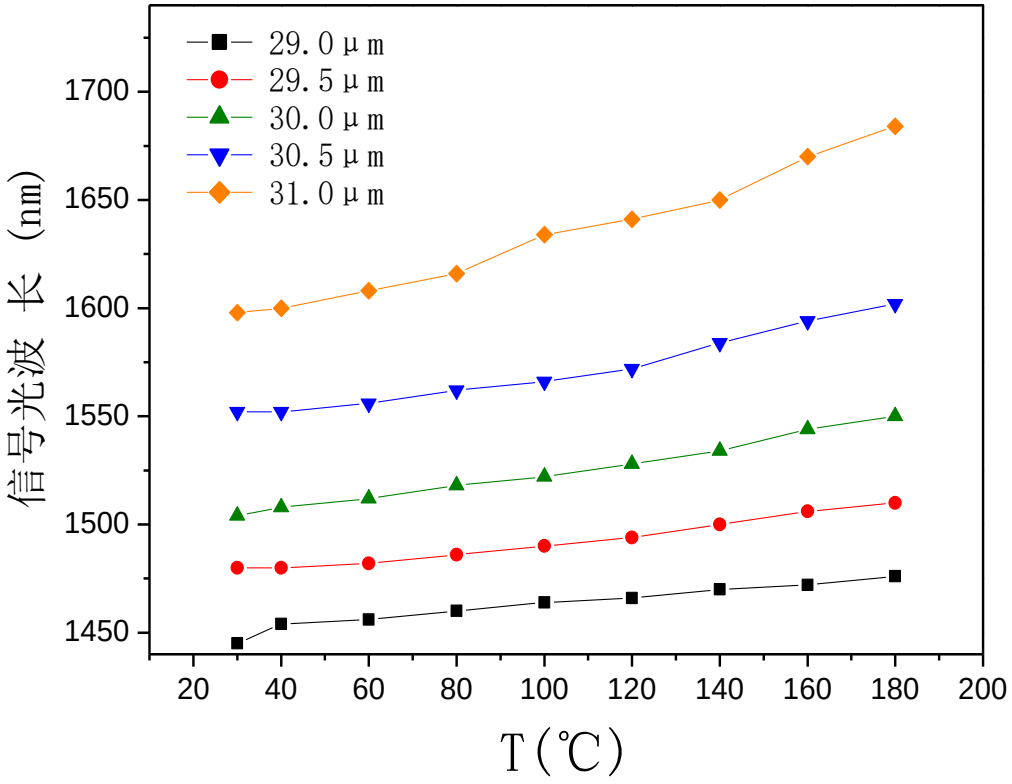
周期极化铌酸锂晶体







PPMgLN OPO周期调谐



29.0μm：1445-1476nm

29.5μm：1478-1510nm

30.0μm：1504-1550nm

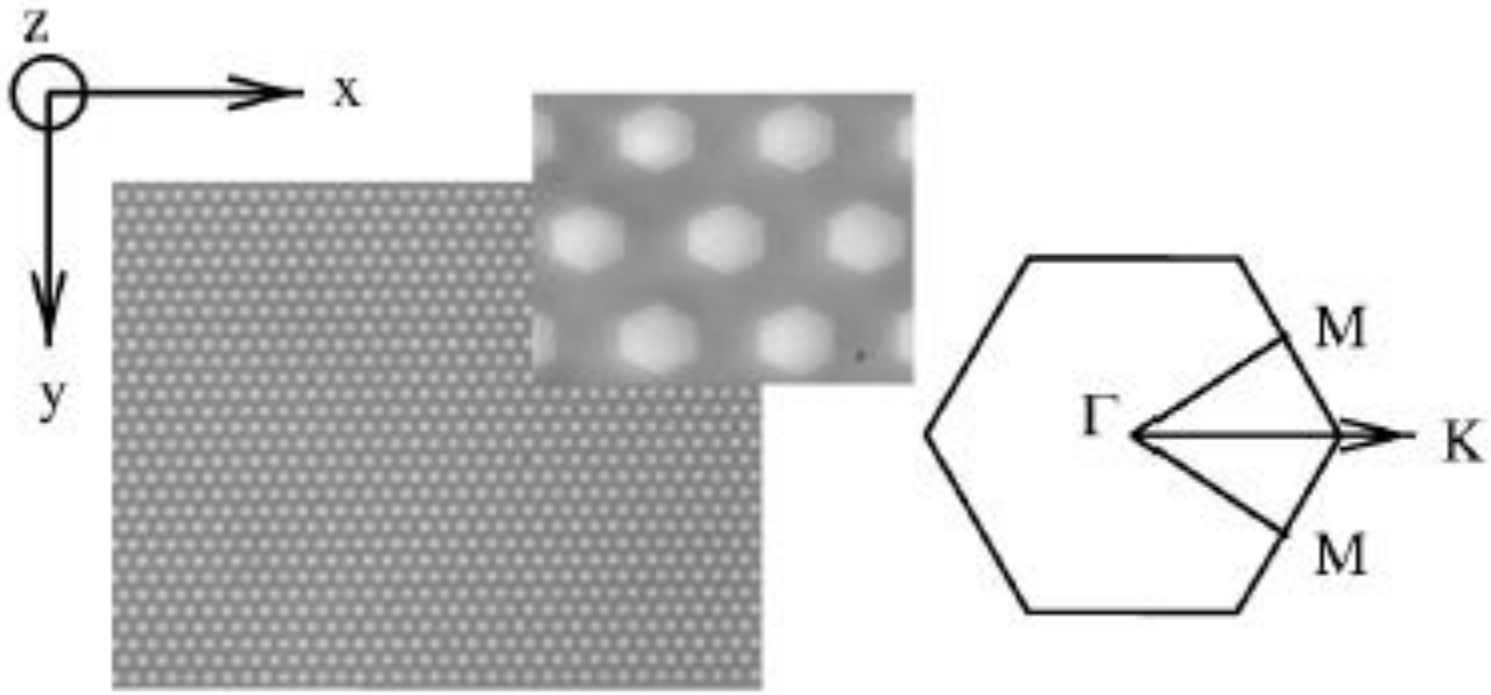
30.5μm：1552-1602nm

31.0μm：1598-1685nm

稳定的连续可调谐宽波段输出： 1445nm-1685nm

不同周期PPMgLN OPO 信号光波长的温度调谐曲线

二维非线性光子晶体

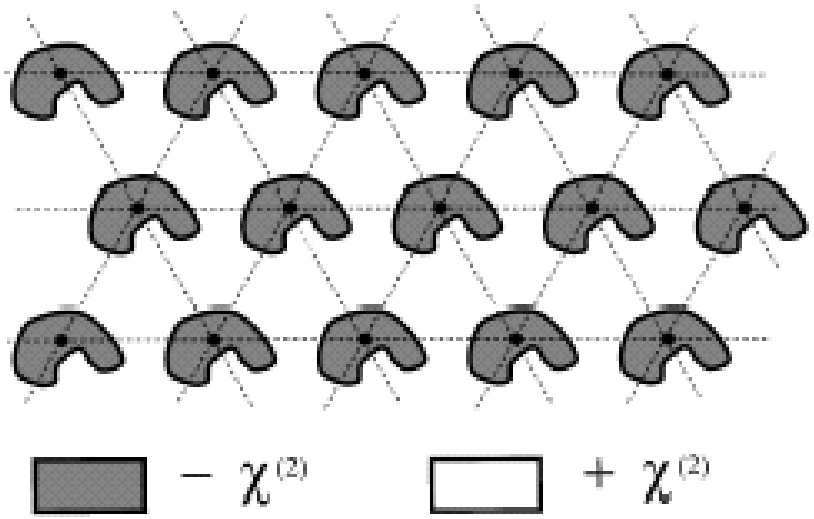


- 二维周期极化铌酸锂晶体

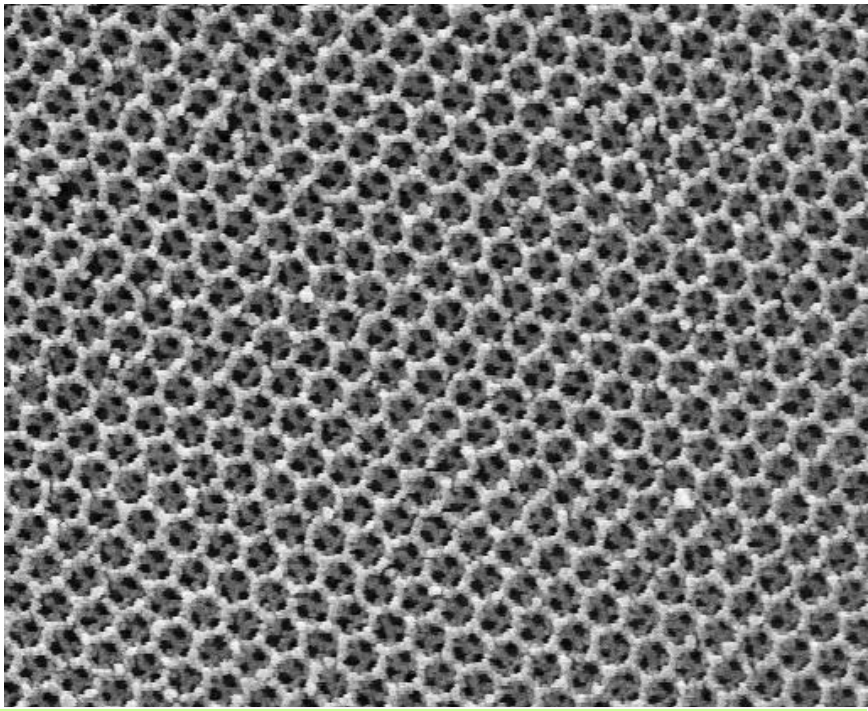
非线性光子晶体—二阶极化率周期性变化

光子晶体—线性极化率$\chi^{(1)}$周期变化—控制光子的运动

非线性光子晶体—非线性极化率$\chi^{(2)}$周期变化—改变光子频率



多波长， 多方向
的高效频率变换



多次填充后的TiO_2反OPAL光子晶体

频率转换中位相匹配

利用晶体的双折射：$2K_o(\omega)=K_e(2\omega)$　　一类 (oo-e)

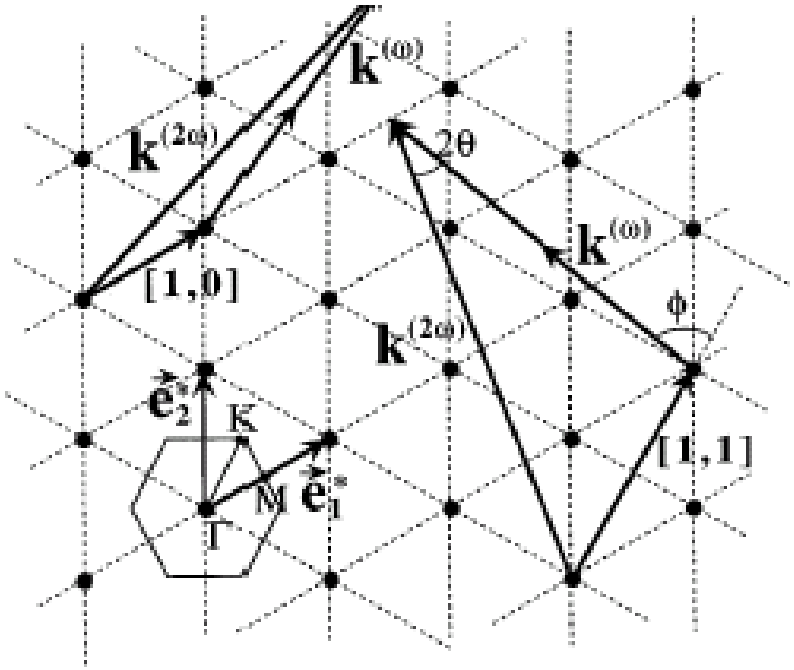
$K_o(\omega)+K_e(\omega)=K_e(2\omega)$　二类 (eo-e)

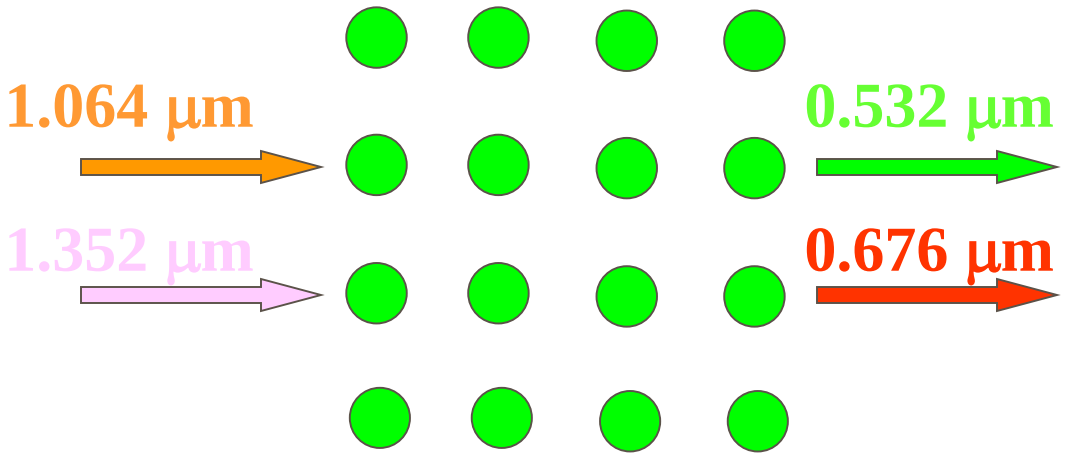
准位相匹配——利用 $\chi^{(2)}$ 周期结构中的倒格失

$$\mathbf{k}^{2\omega}\cdot\nabla\left[E^{2\omega}(\mathbf{r})\right]=-2i\frac{\omega^2}{c^2}(E^{\omega})^2\chi^{(2)}(\mathbf{r})\exp\left[i(\mathbf{k}^{2\omega}-2\mathbf{k}^{\omega})\mathbf{r}\right].$$

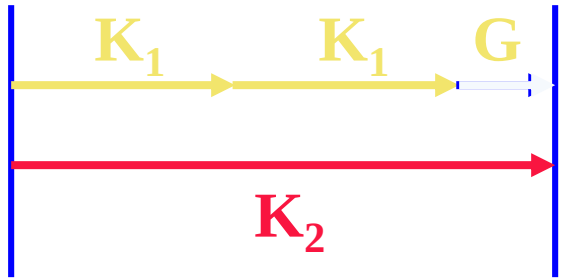
$$\chi^{(2)}(\mathbf{r})=\sum_{\mathbf{G}\in RL}\kappa_{\mathbf{G}}\cdot\exp(-i\mathbf{G}\cdot\mathbf{r})$$

$$\mathbf{k}^{2\omega}-2\mathbf{k}^{\omega}-\mathbf{G}=\mathbf{0}.$$

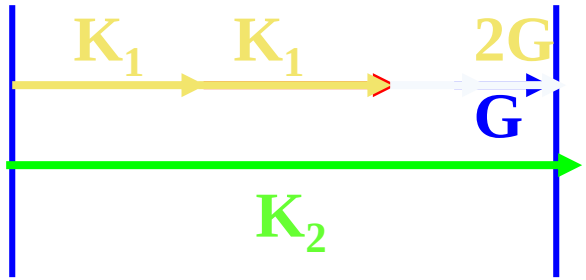




激光在晶体里的走向示意图

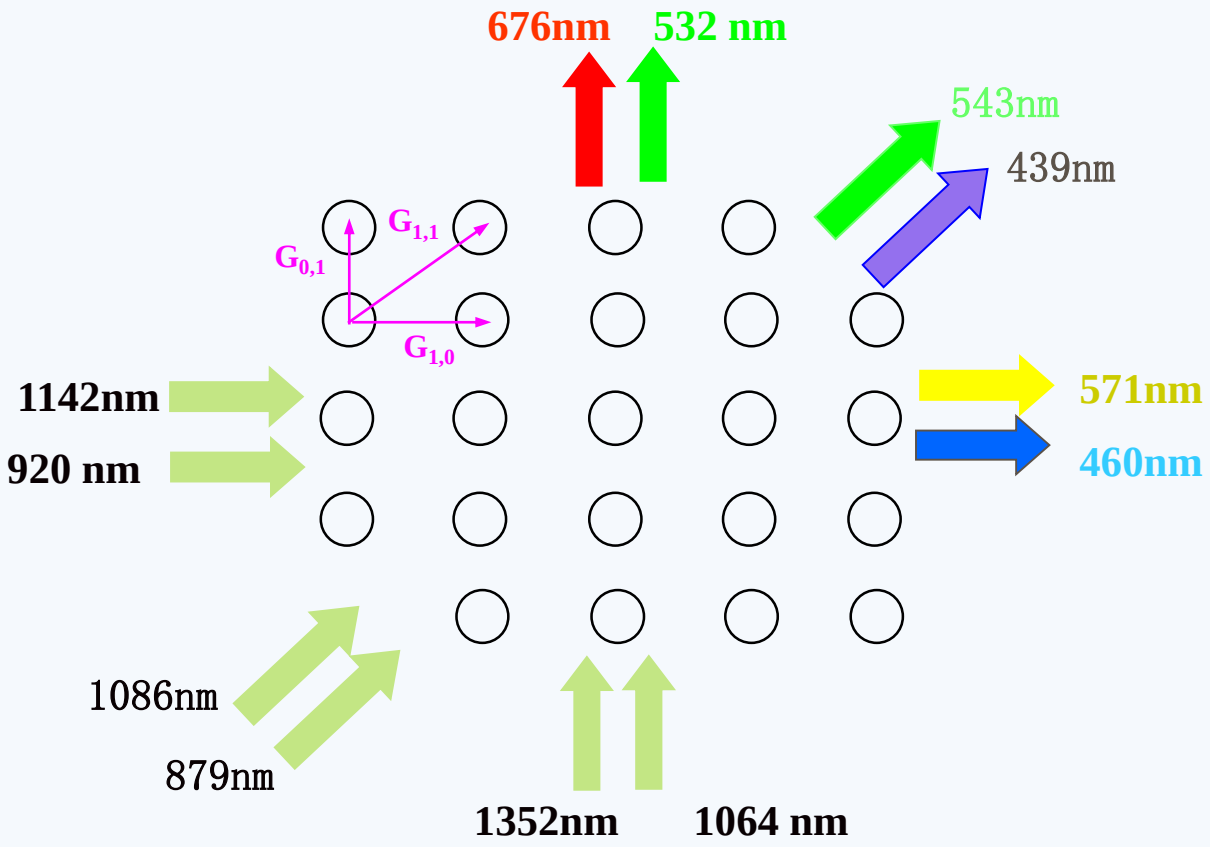


一阶准相位匹配示意图



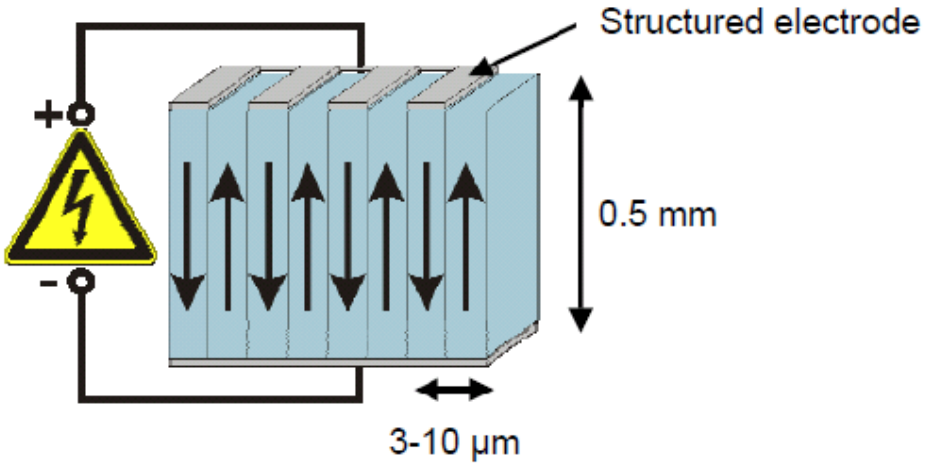
二阶准相位匹配示意图

G为PPLN的倒格矢

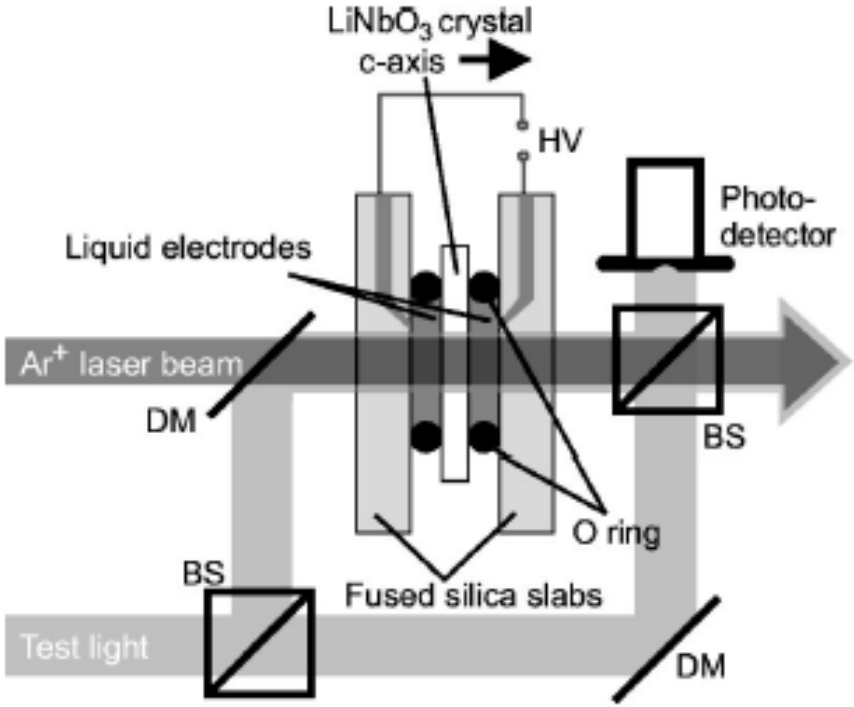


在一块PPLN晶体中利用一阶和二阶准相位匹配实现六个波长倍频

光诱导畴反转



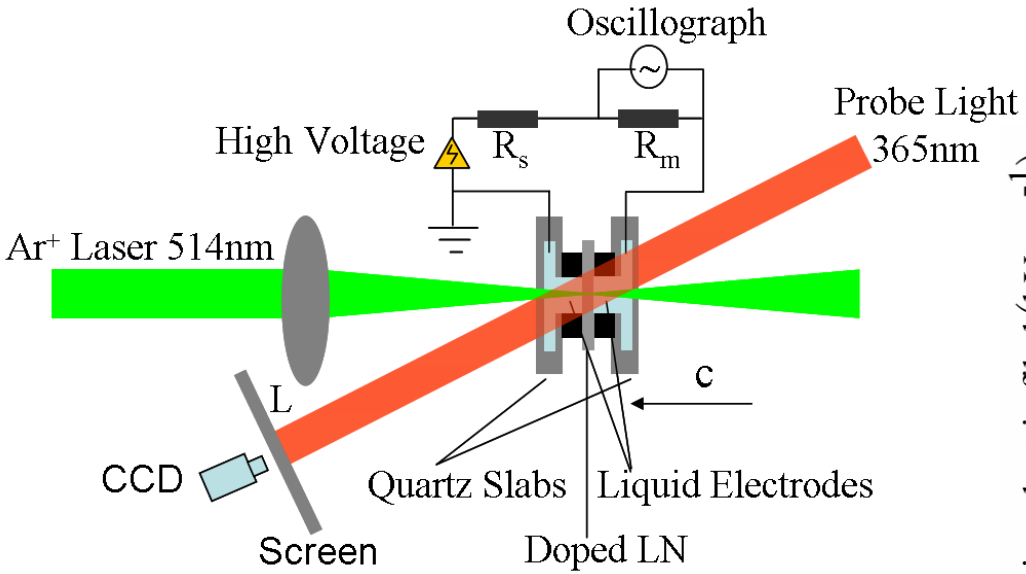
- Poling directly by external Electric field



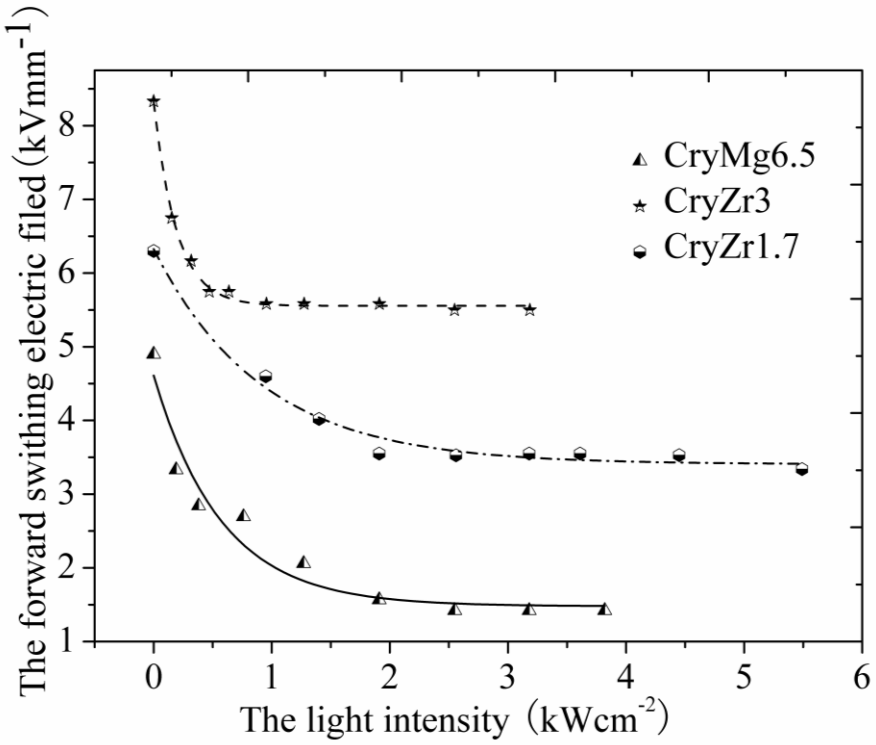
- Set-up for light-induced domain reversal

V. Dierolf et al., *Appl. Phys. Lett.* 84, 3987 (2004)
C. L. Sones et al., *Appl. Phys. Lett.* 86, 212901 (2005)

Light-induced domain reversal of doped LN crystals



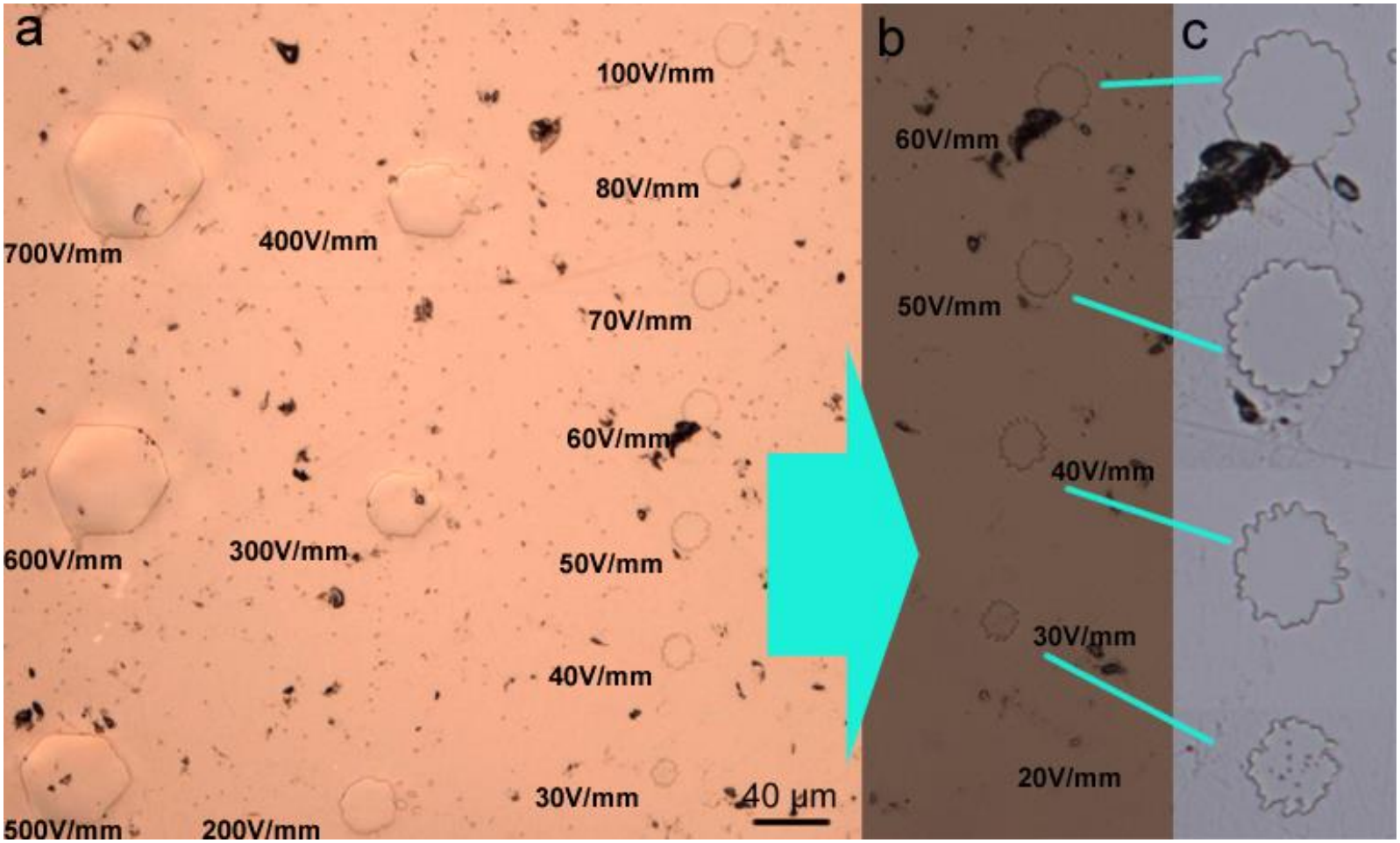
- Set-up for light-induced domain reversal



- Relationship between the reverse field and light intensity.

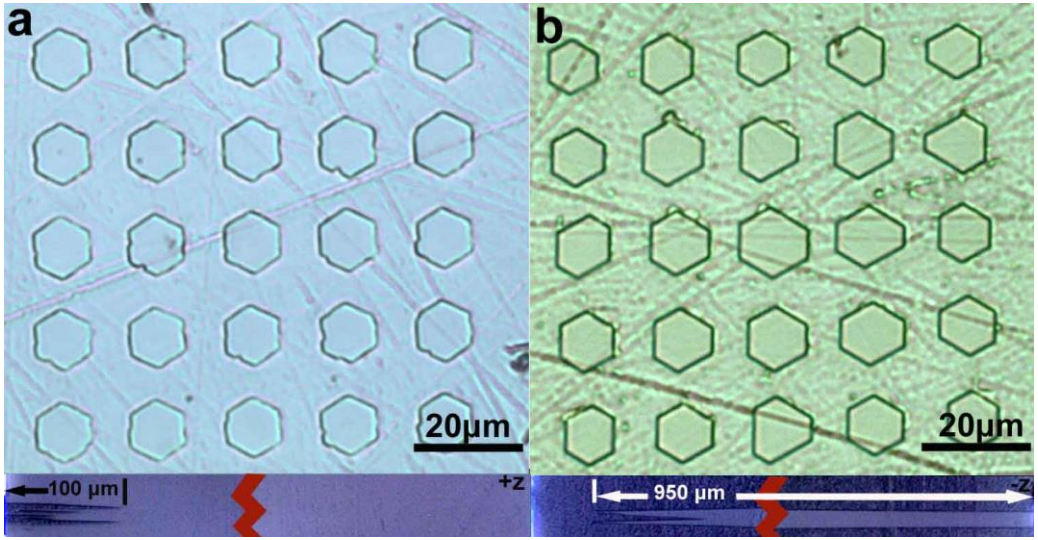
W. Wang et al., J. *Appl. Phys. 105, 043105* (2008)

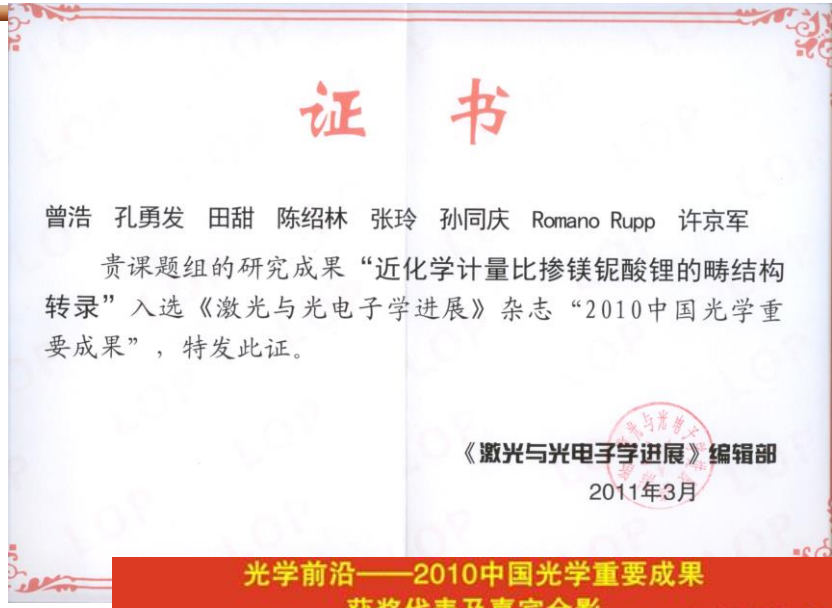
Light-induced superlow electric field for domain reversal



H. Zeng et al., *J. Appl. Phys.* 107, 063514 (2010)

Transcription of domain patterns





证 书

曾浩 孔勇发 田甜 陈绍林 张玲 孙同庆 Romano Rupp 许京军

贵课题组的研究成果“近化学计量比掺镁铌酸锂的畴结构转录”入选《激光与光电子学进展》杂志“2010中国光学重要成果”，特发此证。

《激光与光电子学进展》编辑部

2011年3月



Comparison between light-induced –z patterns and transcribed +z patterns. (a) Two dimensional domain patterns on –z face fabricated by a focal laser beam and electric field of 600 V/mm and a duration time of 10 s. (b) The corresponding transcribed domain patterns on +z face. The imagines beneath are etched y face profiles of domain spots.

H. Zeng et al., *Appl. Phys. Lett.* 97, 201901 (2010)

五、近化学计量比晶体

同成分晶体的缺点：

- 带来了大量的本征缺陷：高达1 mol%的反位铌(Nb_{Li})和4 mol%的锂空位(V_{Li})
- 大量的小极化子、双极化子、局域钛铁矿结构等缺陷。
- 越来越多的实验结果表明，如此众多的本征缺陷限制了晶体的诸多性能。

- 近期，**近化学计量比铌酸锂晶体**以其优异的物理性能引起了人们的极大兴趣。

Characteristics	CP $LiNbO_3$	NS $LiNbO_3$
Absorption edge (nm, α=20 cm^{-1})	319	303 (c_c=50.0)
OH absorption spectrum (cm^{-1})	peak at 3483; half width about 30	Band peak at 3466 half with 3 (c_c=50.0)
Raman spectrum	11 intrinsic modes; HW of 153 is 10.7	13 intrinsic modes (c_c=49.74); HW of 153 is 5.8 (c_c=49.95)
EPR spectrum linewidth (mT, Fe^{3+}, 14K, 300 mT)	45	5 (c_c=49.95)
Birefringence index (633 nm)	–0.0837	–0.0974 (c_c=49.74)
Electro-optic coefficient (pm V^{-1})	r_{61}= 6.07	r_{61}=9.89 (c_c=49.95)
Nonlinear coefficient (pm/V)	13.8	17.5 (c_c=49.8)
Saturation values of photo-induced birefringence change	1×10^{-5}	10×10^{-5} (c_c=49.8)
Response time of PR (s)	several hundreds	0.2 (c_c=49.9)
The sensitivity of nonvolatile holographic storage	1.7×10^{-4}	3×10^{-2} (c_c=49.65)
Resistance to light induced optical damage (kW/cm^2)	10^0	10^4 (c_c=49.5~48.2, and 1.8 mol % MgO doped)
Domain reversion V (kV/mm)	21	0.2 (c_c=50.0)
Emission intensity (arb. Units)	0.47	17 (c_c=50.0±0.15)

2001 CTA Awards

- the winners of the *Laser Focus World* Commercial Technology Achievement Awards for 2001.

Laser Focus World

January, 2001

Valerie Coffey



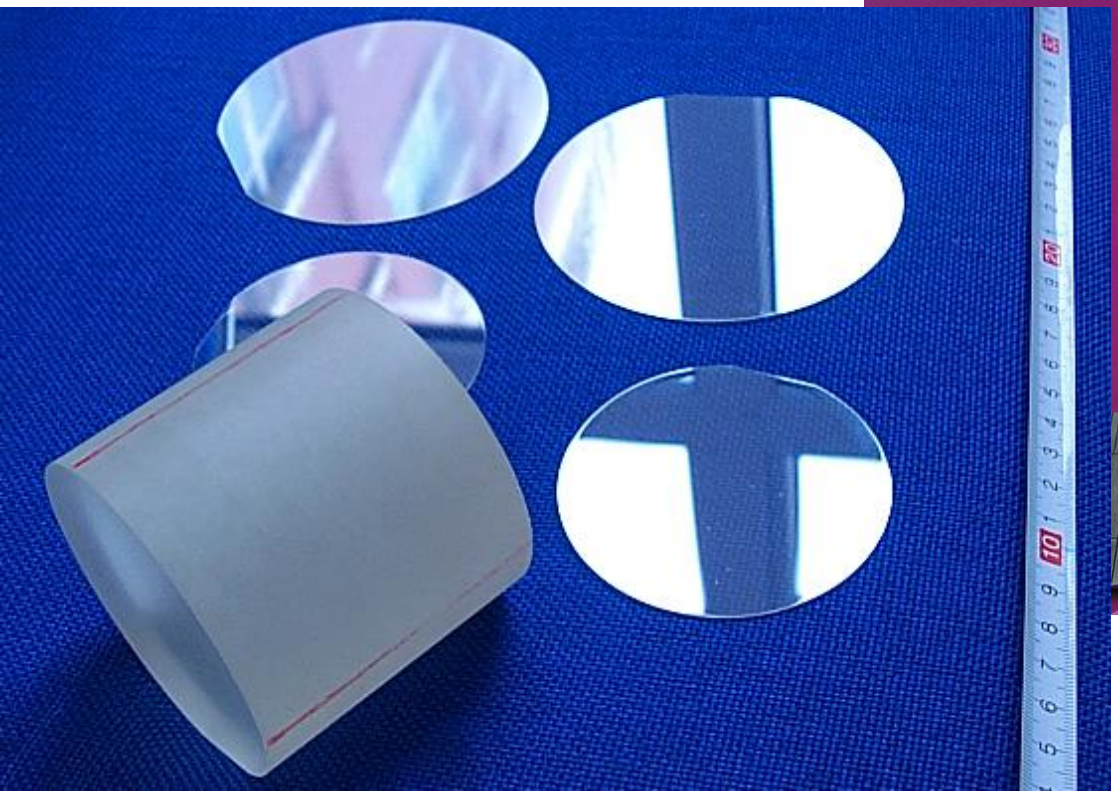
magnesium-doped stoichiometric lithium niobate (Mg:SLN)

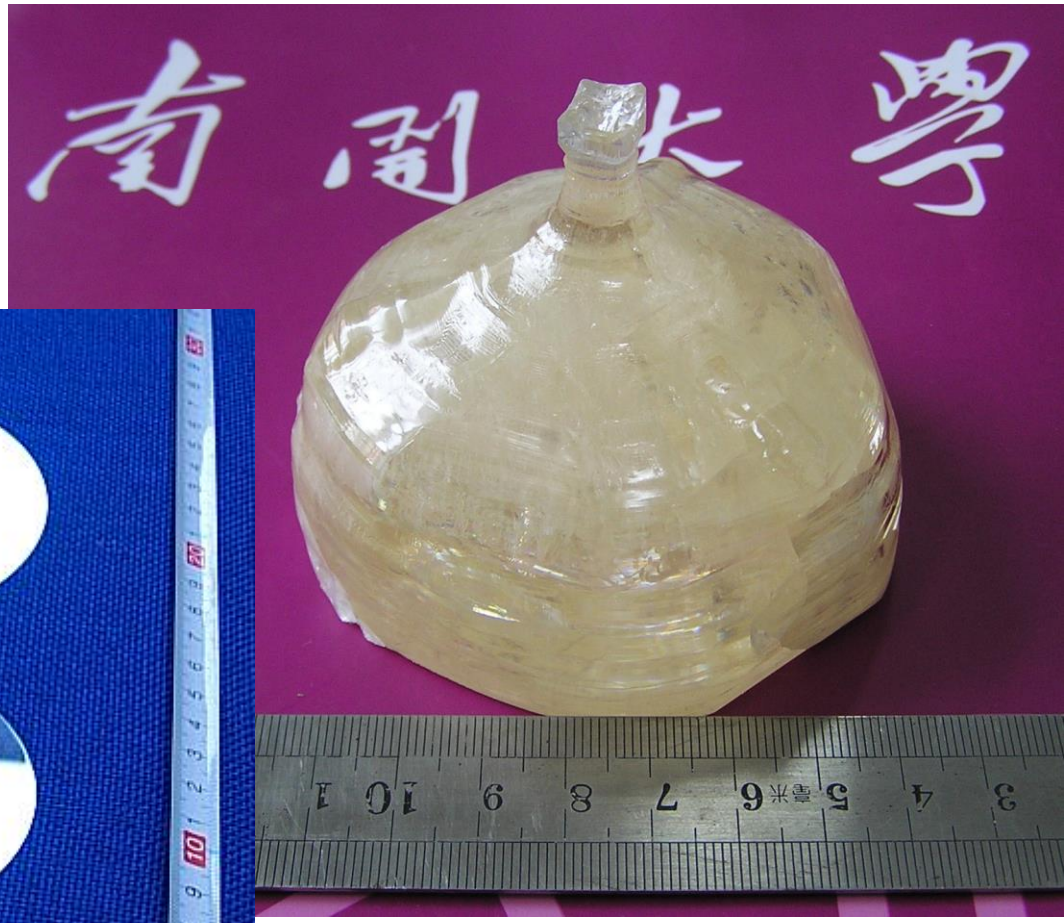
OPTICS/OPTICAL MATERIALS

Deltronic Crystal Industries Inc.

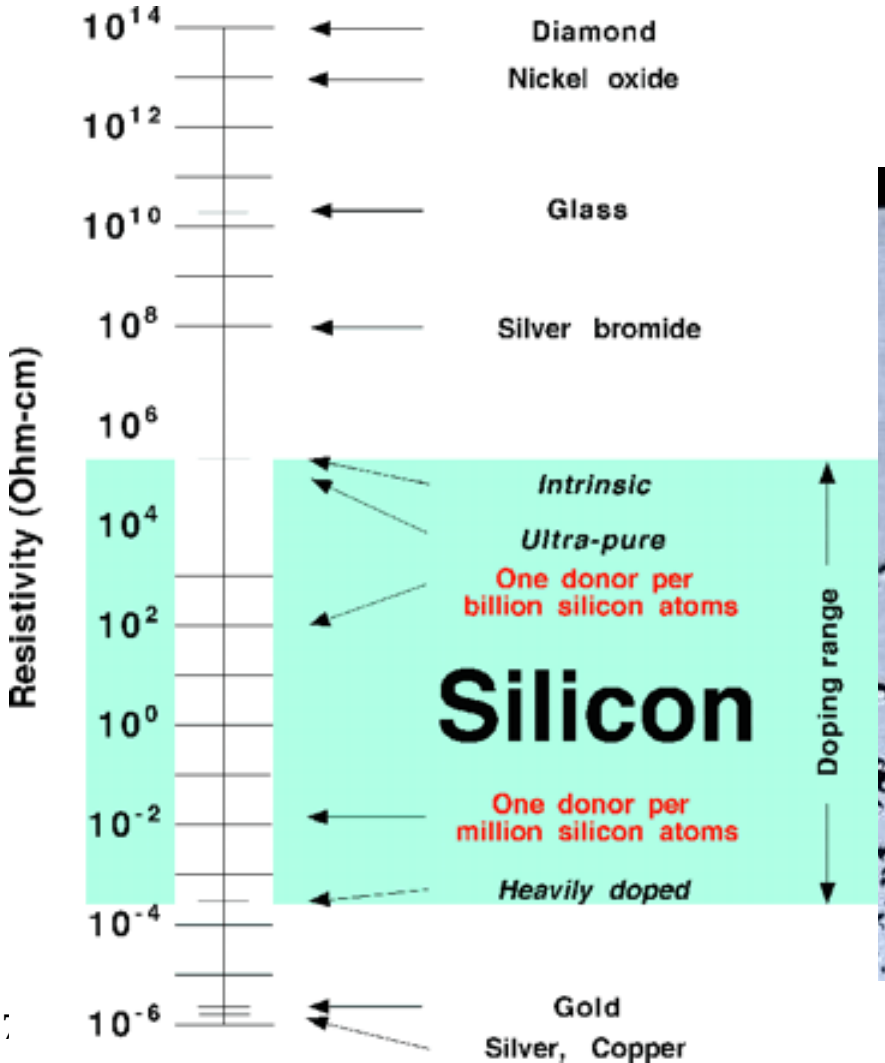
(Dover, NJ)

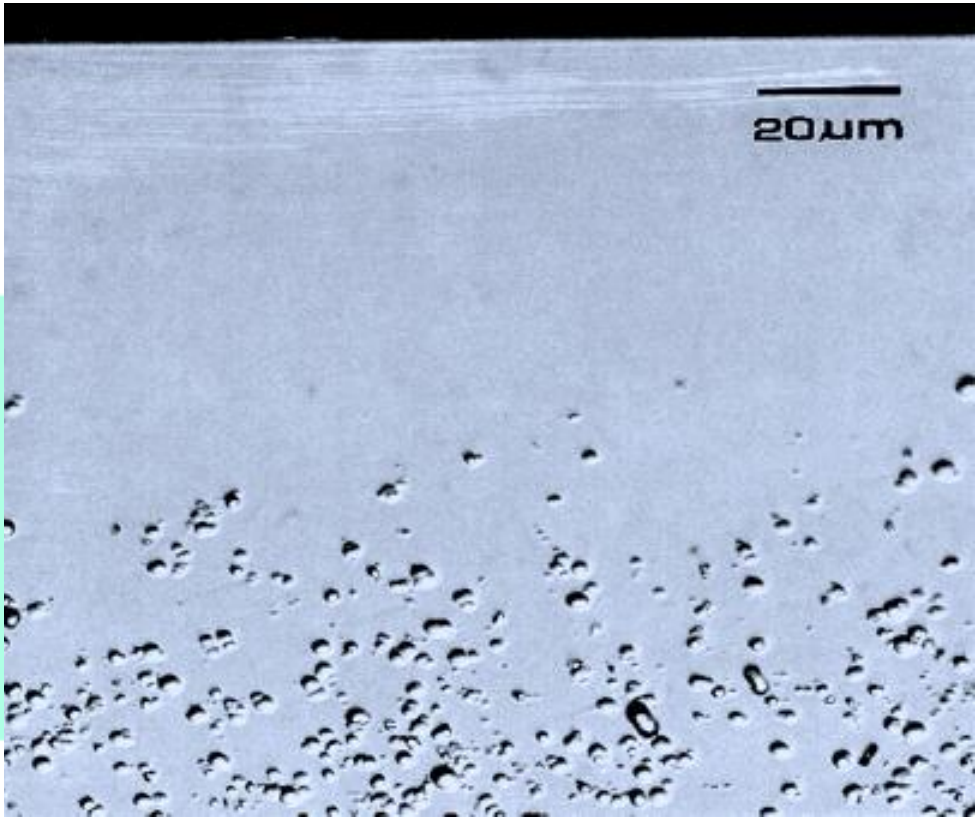
近化学计量比铌酸锂晶体





单晶硅


Resistivity (Ohm-cm)
10^{14}
10^{12}
10^{10}
10^{8}
10^{6}
10^{4}
10^{2}
10^{0}
10^{-2}
10^{-4}
10^{-6}
Diamond
Nickel oxide
Glass
Silver bromide
Intrinsic
Ultra-pure
One donor per billion silicon atoms
Silicon
One donor per million silicon atoms
Heavily doped
Doping range
Gold
Silver, Copper


20μm

石英光纤



- 1966年高锟和Hockham提出损耗为20dB/km的光导纤科用于通讯时，世界上最优良的光学玻璃的衰减为 1000 dB/km 。
- 1970年，Corning公司首先制成衰减为20dB/km的光纤。
- 1974年，光纤衰减降低到2dB/km (原料纯度提高到8N) 。
- 1976年，光纤衰减降低到0.5dB/km（原料中OH离子的浓度控制在ppm量级）。
- 1980年，传输损耗仅有 0.2 dB/km （接近理论值0.15dB/km）。



近完美铌酸锂晶体

- 近化学计量比使铌酸锂晶体性能的突破及单晶硅和石英光纤的巨大成功令我们深思：

  如果铌酸锂晶体的本征缺陷浓度能减小到10^{-5}、10^{-6}或者10^{-8}，如果铌酸锂晶体的生长原料的纯度能提高到5N、6N、甚至8N，如果铌酸锂晶体的缺陷浓度能降低到100/mm^{2}以下，

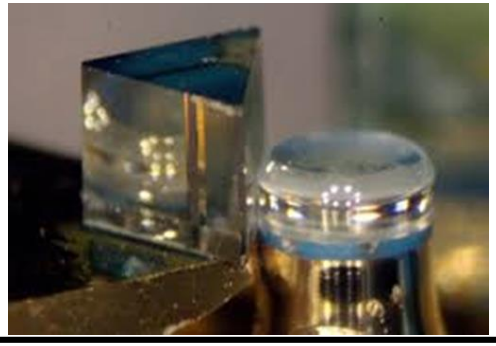
我们会发现什么样的实验现象和物理问题？铌酸锂晶体的性能会怎样呢？

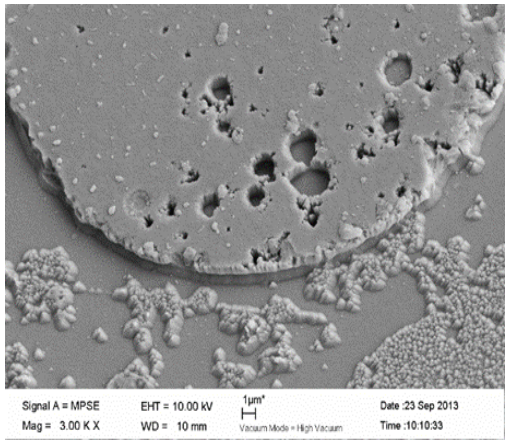
六、结语

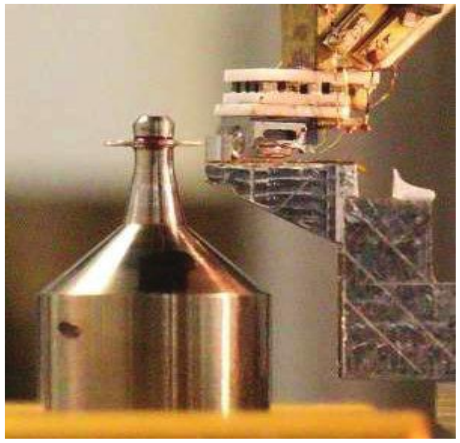
机遇： 具备宽透明窗口、高非线性系数、低吸收损耗，并具有高电光、热光以及声光系数。

➢ 全能冠军，“光学硅”、“非线性光学硅”、“光子学硅”。

	二阶非线性	电光系数
铌酸锂	**41.7 pm/V**	**30.9 pm/V**
石 英	0.335 pm/V	0.2 pm/V



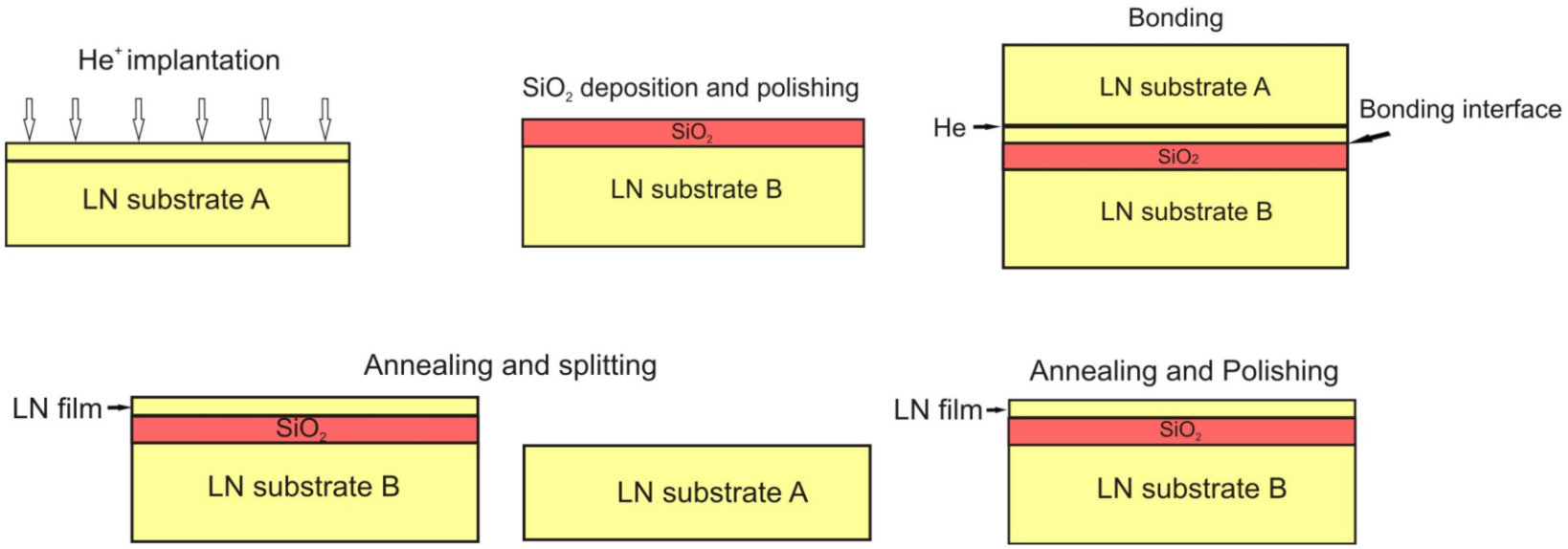


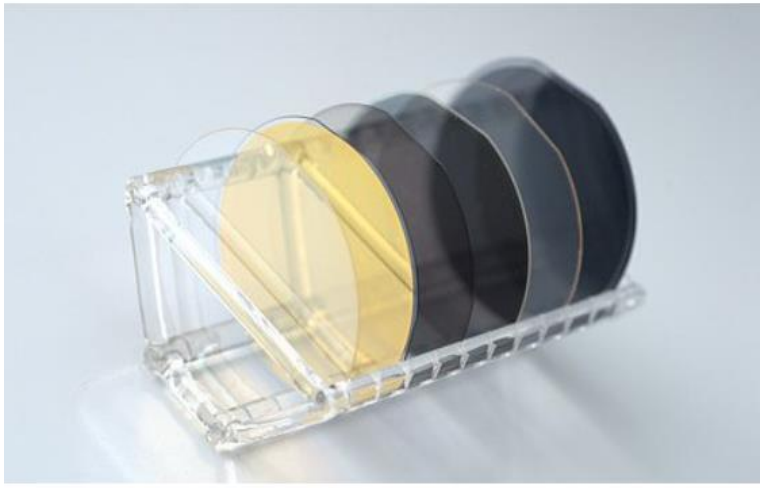


挑战：

- 化学性质稳定、耐腐蚀、机械研磨效率低
- 体快单晶难以兼容半导体微纳加工工艺，阻碍其芯片化、集成化

铌酸锂薄膜 (Lithium Niobate On Insulator, LNOI)

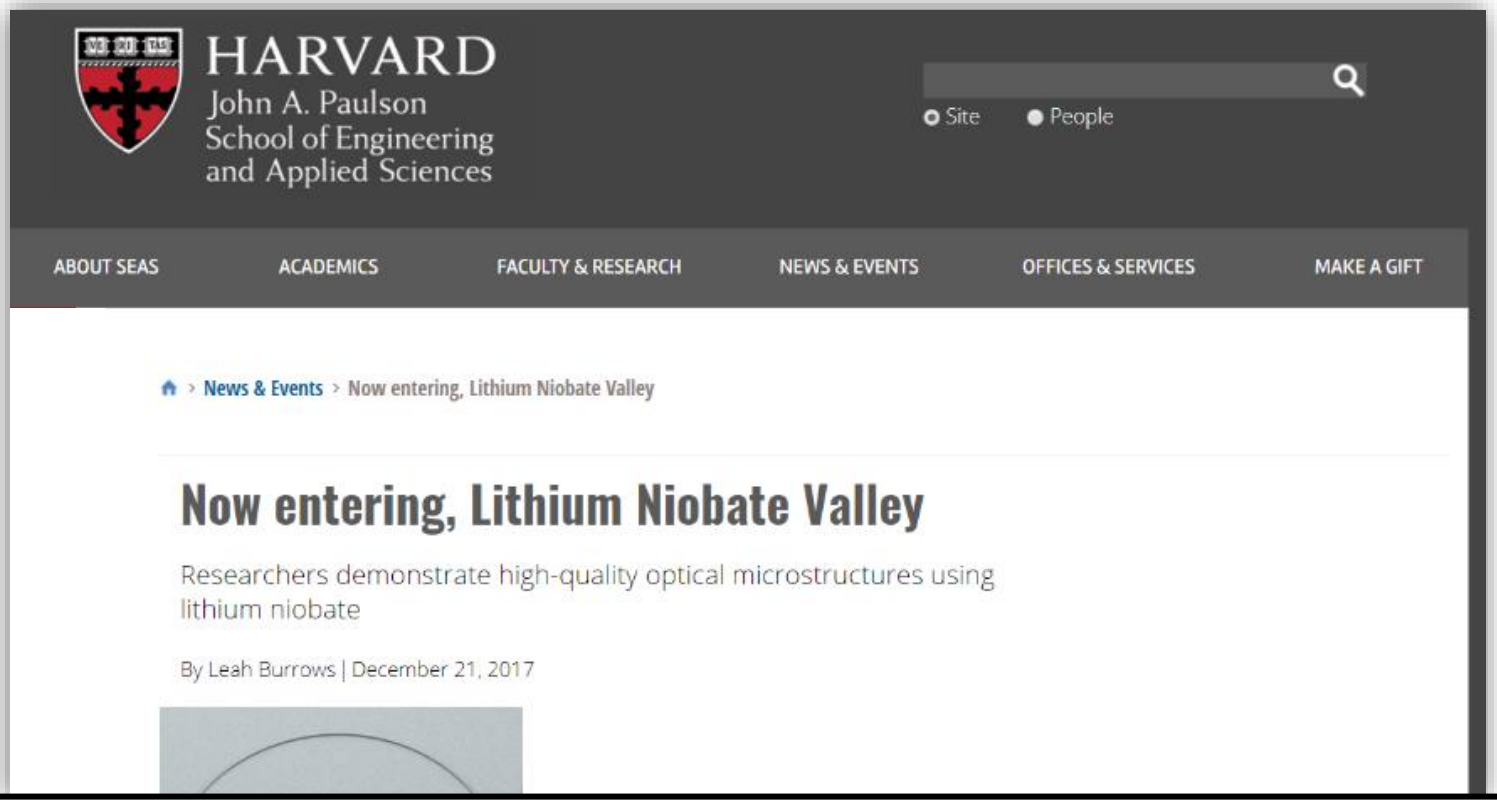




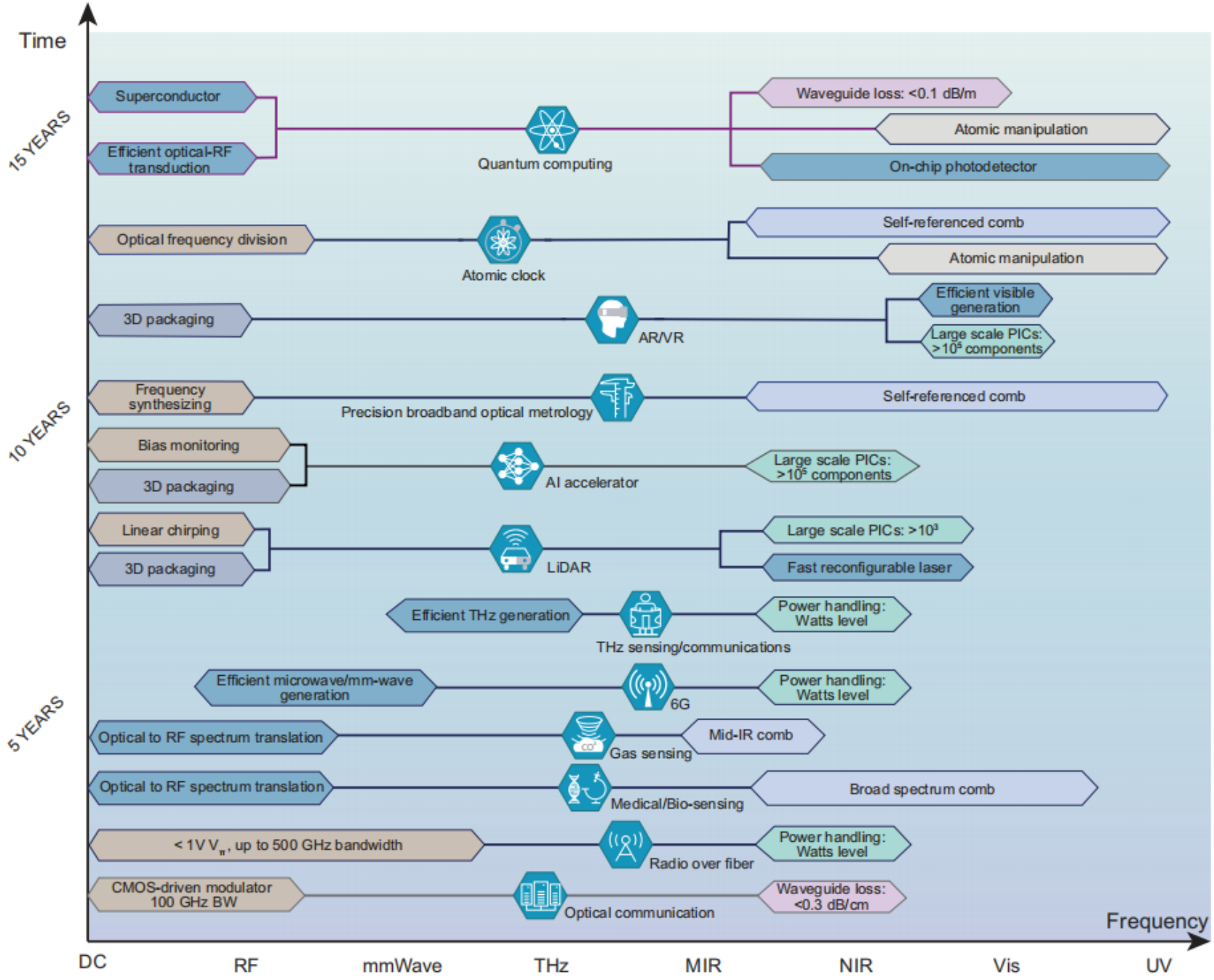
优势：

- 更高的折射率梯度
- 更强的光场局域性
- 兼容半导体微纳加工工艺

人类正在进入“铌酸锂谷”时代



如果集成电子学的核心是“硅”材料，那么光子学革命应该是属于“铌酸锂”的。铌酸锂对于光子学的意义，等同于硅材料对于电子学的意义。



Outlook of LN photonics and its applications.

Science 379, eabj4396 (2023).

为表彰天津市自然科学奖获得者，特颁发此证书。

项目名称：三维光折变存储材料及其存储器的研究

奖励等级：一等

获 奖 者：南开大学

天津市自然科学奖

证 书

二〇〇二年四月十三日

奖励编号：ZR2001-1-002-D1

为表彰天津市技术发明奖获得者，特颁发此证书。

项目名称：大尺寸光学级铌酸锂晶体

奖励等级：二 等

获 奖 者：孔勇发

天津市技术发明奖

证 书

二〇〇六年四月十三日

证书编号：2005FM-2-003-R1

为表彰天津市科学技术进步奖获得者，特颁发此证书。

项目名称：系列实用高温度稳定性电光调Q激光系统的开发与应用

奖励等级：一 等

获 奖 者：孔勇发

天津市科学技术进步奖

证 书

二〇一三年一月二十六日

证书编号：2012JB-1-004-R5

国家自然科学奖

证 书

为表彰国家自然科学奖获得者，特颁发此证书。

项目名称：光折变新效应、机理与器件的研究

奖励等级：二等

获 奖 者：许京军(南开大学)

2005年11月20日

证书号：2005-Z-107-2-02-R01

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Yongfa Kong, Fang Bo, Weiwei Wang, Dahuai Zheng, Hongde Liu, Guoquan Zhang, Romano Rupp, and Jingjun Xu**

Adv. Mater. **2020**, *32*, 1806452

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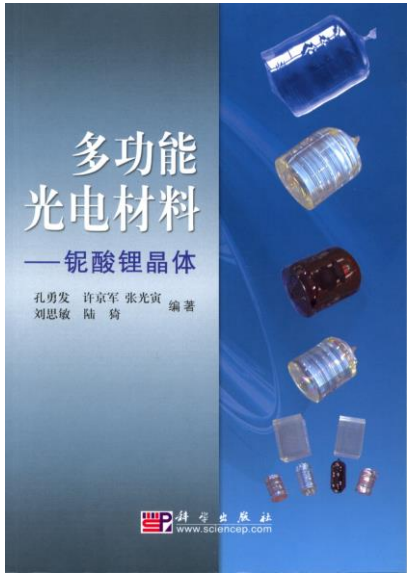
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铌酸锂的荦荦之路:历史与若干进展

高博锋,任梦昕,郑大怀,几 伟,蔡 卫,孙 军,孔勇发,许京军

(南开大学物理科学学院 & 泰达应用物理研究院,弱光非线性光子学教育部重点实验室,天津 300071)

…学特性于一身,被认为是非线性光学的模型晶体,已经表现出巨大的实用价…经在国土安全、医学检测、高能物理、工业探测等领域占据着不可或缺的地…纳结构中新型光学效应的研究,已经成为国际上竞相争夺的前沿热点之…具有重要推动作用。本文主要围绕铌酸锂的光学性质综述了其发展历…,并对其未来发展进行了展望。



欢迎进入铌酸锂谷！



D. Zhu, *et al*. Adv. Opt. Photon. 13, 242 (2021).